

Research Article

Lucio Palazzo*, Riccardo Ievoli and Giancarlo Ragozini

Testing styles of play using triad census distribution: an application to men's football

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Abstract: Summary statistics of football matches such as final score, possession and percentage of completed passes are not satisfyingly informative about style of play seen on the pitch. In this sense, networks and graphs are able to quantify how teams play differently from each others. We study the distribution of triad census, i.e., the distribution of local structures in networks and we show how it is possible to characterize passing networks of football teams. We describe the triadic structure and analyse its distribution under some specific probabilistic assumptions, introducing, in this context, some tests to verify the presence of specific triadic patterns in football data. We firstly run an *omnibus* test against random structure to assess whether observed triadic distribution deviates from randomness. Then, we redesign the Dirichlet-Multinomial test to recognize different triadic behaviours after choosing some reference patterns. The proposed tests are applied to a real dataset regarding 288 matches in the Group Stage of UEFA Champions League among three consecutive seasons.

Keywords: count data; multivariate testing; passing network; sport performance; triad census.

1 Introduction

In past decades, decision-makers in sports used to be subjective, trusting mainly in intuition and personal experience rather than evidences from the data. Nowadays, sport analytics and availability of data facilitate the usage of

statistical methodologies in a broad range of sports (such as basket, cricket, hockey and golf) recently spreading even in football (for a wide review of the literature see Kimber and Hansford 1993; Cox and Trevor 2002; Albert, Bennett, and Cochran 2005; Lewis 2005; McIntyre and McKittrick 2005; Sampaio et al. 2006; Ibáñez et al. 2008; Tibshirani, Price, and Taylor 2011; Moura, Martins, and Cunha 2014, and references therein).

A relevant part of the literature has been focused on player-level analysis, see, e.g., Kahn (1993); Baumer, Jensen, and Matthews (2015) (baseball), Fewell et al. (2012) (basketball), Hadley et al. (2000) and Yurko, Ventura, and Horowitz (2019) (American football), Gomez (2002), (hockey) and Gallagher, Frisoli, and Luby (2021) (tennis). For what concerns football, increasingly sophisticated statistical models are used to predict football outcomes (Groll, Schauburger, and Tutz 2015), even considering players abilities (Carpita, Ciavolino, and Pasca 2019), or teams' final rankings (Groll et al. 2019).

At a team-level perspective, tactics and strategies are key elements for success in football. Statistical methodologies that are used in this context are still under debate: conventional approaches are based on the analysis of ball possession (Bate 1988; Lago-Peñas and Dellal 2010), while others rely on summary statistics (Clemente et al. 2016) or on the prediction of the probability of scoring goals (Keller 1994; Stern 1991).

In recent years network analysis has been applied on football passing distributions. A seminal contribution can be found in Grund (2012), helping to establish a relationship between network structure (measured by “intensity”, i.e., passes over minutes, and centralization-based indices) and team performance (measured in terms of scored goals). More recently, a set of network summary measures from the passing distribution has been used to model the probability of win the game (Ievoli, Gardini, and Palazzo 2021a), while passing network indicators have been involved in Bayesian Hierarchical models (using regularization methods) to model the scored goals or the difference in goals (Ievoli, Gardini, and Palazzo 2021b). Other interesting applications of network analysis in football can be found in Pina, Paulo, and Araújo (2017), Clemente, Sarmiento, and

*Corresponding author: **Lucio Palazzo**, Department of Political Sciences, University of Naples Federico II, Napoli, Italy, E-mail: lucio.palazzo@unina.it. <https://orcid.org/0000-0001-7529-4689>

Riccardo Ievoli, Department of Chemical, Pharmaceutical and Agricultural Sciences, University of Ferrara, Ferrara, Italy. <https://orcid.org/0000-0001-9489-3564>

Giancarlo Ragozini, Department of Political Sciences, University of Naples Federico II, Napoli, Italy

Aquino (2020), Mclean et al. (2018), and Gonçalves et al. (2017).

Starting from the aforementioned literature, we propose a network analysis approach based on triad census to extract the properties of passing behaviour in football that should be informative to identify and characterize team strategies.

We focus on passes since they are undoubtedly the most representative events in football, especially compared to goals, shots and other in-field variables (Cintia, Rinzivillo, and Pappalardo 2015). Although the literature has been mainly focused on scored (and expected) goals, recent studies highlight the importance of passing effectiveness (Rein, Raabe, and Memmert 2017) and passing behaviours in evaluating overall players' performance (Bransen, Van Haaren, and van de Velden 2019). Bransen, Van Haaren, and van de Velden (2019) also evidence that passing distribution among players is able to characterize players' profiles, also allowing to make comparisons between them. Indeed, starting from the (undirected) passing networks of the Italian teams in first division ("Serie A"), Diquigiovanni and Scarpa (2019) identify 15 different tactics using a two phase clustering technique.

Passes are not only related to individual technical aspects or "vision" throughout the pitch, but also express social attitudes of players which may include "*sympathies and antipathies, good versus bad cooperation partners, kindness versus resentment*" (Hyballa and Te Poel 2015, p. 55). In this sense, each social phenomenon expressing a particular structure relies on properties such as density, transitivity and presence of clusters (Faust 2006). Then triad census, widely surveyed by Faust (2010), appears a suitable method to take into account all of these features.

Our aim is twofold: we want to study the triadic distribution in football passing networks using the resulting information to test if it significantly deviates from a random scenario. We also formally test for the presence of one or different styles of play in football according to a set of reference "strategies" determined by specific triadic configurations.

To pursue the research aim, at first we adopt the *omnibus* test for the observed triadic distribution against a random structure. The test is based on inferential and probabilistic properties of triads, discussed in Holland and Leinhardt (1978) and Faust (2010), and can be useful to find how observed triads deviate from randomness. However, testing only the non-randomness of triads does not allow to identify different styles of play that could be developed in a game, expressed through the distribution of passes. Thus, the omnibus test is not able to provide comparisons at team level.

To overcome this issue, we redesign a multivariate test based on Dirichlet-Multinomial (DM) distribution. This family of tests have been developed and applied within several research contexts (cf. Ennis and Bi 1999; La Rosa et al. 2012; Ricard and Davison 2007; Wu et al. 2017, among others), and it is suitable in case of non-independent counts. However, to the best of our knowledge, the DM-based tests have never been applied to triad census, especially for football data.

The main contribution of this work is to exploit a more flexible testing approach to triad census with the purpose to unveil the potential of triadic distribution in football passing network analysis, especially to retrieve possible styles of play.

The paper is organized as follows: Section 2 introduces the main concepts and theory of team passing networks and triad census, while in Section 3 the theory behind the *omnibus* test and the DM-based test is introduced. Section 4 includes an application regarding three European Champions League Seasons from 2016 to 2017 to 2018–2019. Finally, in Section 5 we highlight the possible main implications of this approach in football along with some conclusions and possible advances.

2 Methodology

Football matches are characterized by complex and multidimensional features: scored points (goals) have very low frequencies with respect to other relevant and detectable events, such as shots, cards, fouls and free kicks among others (Cintia, Rinzivillo, and Pappalardo 2015). Nowadays, new technologies such as image analysis, wearable devices, multiple-camera player trackers and drone-based analysis of training sessions, are changing the way to retrieve data and provide new opportunities in football tactics (cf. Boyle and Haynes 2004; Buchheit and Simpson 2017; Buchheit et al. 2014; Edgecomb and Norton 2006). Among them, ball passes between players of the same team are complex events to analyse and their study can be crucial to retrieve useful information regarding underlying mechanics determining style of play and the overall team performance. A first descriptive approach is given by the analysis of type and frequency of passes between players in a team, during each match.

Resulting data share a "relational" structure, defined as a set of "measurements" between pairs of units, also called "agents" and may involve contacts, connections, group attachments, meetings or relations between agents and these links, expressed by ties binding individuals together. The overall structure coming from football matches, defined as team passing network, can be described as the relationship between player (agents) in terms of passes (measurements) in a valued square matrix. Within this framework, comparisons at individual (micro) level involving players are generally straightforward and some global information can be also extracted in a more general context, in order to evaluate performance of teams in a macro level perspective. A systematic literature review on the relationship between passing network and tactic analysis in football can be found in Caicedo-Parada, Lago-Peñas, and Ortega-Toro (2020).

2.1 Passing networks in football

A passing network G is defined as the ordered triple $(\mathcal{V}, \mathcal{A}, \mathcal{W})$ consisting of a set of 11 first team players $v \in \mathcal{V}$, a set of arcs $a \in \mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$, consisting of all the passing relationships between them and a set of weights $\omega \in \mathcal{W}$, i.e., the number of passes occurred during the match, denoting the strength of the connection. We assume that both vertices $\mathcal{V}$ and arcs $\mathcal{A}$ are finite as well as $|\mathcal{V}| = n$. Passing network can be expressed in the form of an adjacency matrix $P = (p_{i,j})$, with $p_{i,j} = \omega(a_{i,j})$ if $\exists a_{i,j} \in \mathcal{A}$ and $p_{i,j} = 0$ otherwise.

In Table 1 we present the team passing network of a match involving Arsenal against PSG in September 13th 2016 (first match of the Group Stage, group A). For example, during the match, goalkeeper Ospina passed successfully the ball to Monreal four times (p_7) and 11 to Mustafi (p_9), while he received once the ball from Bellerin (p_{10}). Figure 1 represents the directed and weighted graph representation obtained through the adjacency matrix of Table 1; eleven agents (representing the players) are connected between arrows and the strength of relationship, i.e., the number of passes, is expressed through the size of them.

We need to remark that values on the main diagonal in Table 1 are constrained to be null since we assume that a player cannot pass the ball to himself. Thus the so-called “loops” are not allowed in this framework.

To summarize, passing networks in football share some peculiarities: they are not high dimensional, involving a small and generally fixed number of vertices, but may also present a certain degree of sparseness. In fact, the number of interactions may depend on (i) position on the pitch (ii) individual skills and team cooperation attitudes and, finally, (iii) team strategies and/or style of play.

In order to formally uncover the qualitative attributes of these links, the analysis of triads represents a possible way to detect patterns in the passing distribution and, under some mild assumptions, make a proper inference. Thus, observed triadic distribution can be used to compare connectivity of a team during competitions, in order to analyse changes in strategies and styles of play. In addition, the comparison could be also made through teams in the same tournament, helping to understand if different passing behaviours can have an impact on the football outcomes.

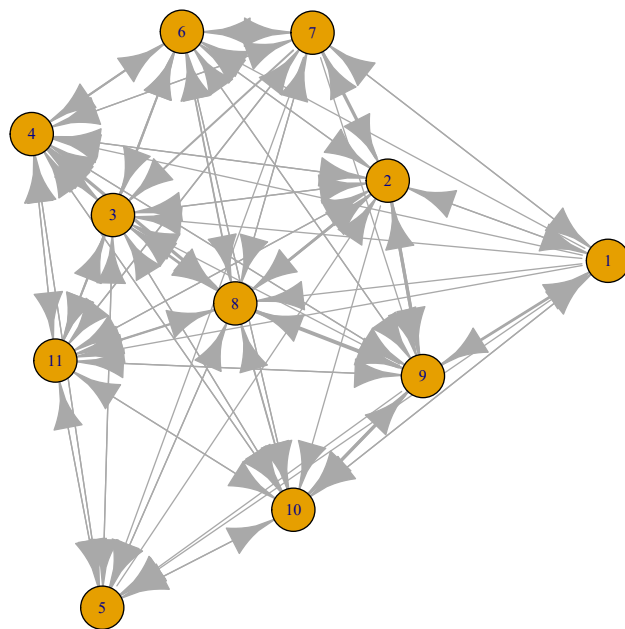


Figure 1: Graph of adjacency matrix of the team passing network of Arsenal illustrated in Table 1.

2.2 Triad census

Although the minimal structural element in a network is a dyad, defined as the interaction occurring between two agents, the minimal group is represented by a triad, involving a subset composed by three agents. In general, given a set of n nodes (agents), a k -subgraph is defined as the set containing all the possible subsets of dimension equal to k of a network. The number of possible distinct k -subgraph that can be examined are $\binom{n}{k}$, with $k \leq n$, and the number of possible unweighted and directed arcs occurring between k individuals is $k(k-1)$, spanning from the totally disconnected (*null*) to the full connected

Table 1: An example of adjacency matrix in football representing the team passing network of Arsenal, first Group Stage match of 2016–2017 Champions League.

	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8	p_9	p_{10}	p_{11}
Ospina	p_1		1	1	1	1	4	1	11	2	1
Koscielny	p_2	1		3	2	0	1	13	11	12	1
Alexis Sanchez	p_3	0	3		9	4	4	0	4	0	4
Ozil	p_4	0	2	8		2	8	3	6	2	3
Oxlade-Chamberlain	p_5	0	2	2	2		0	0	4	0	2
Iwobi	p_6	0	4	6	8	0		7	3	4	4
Monreal	p_7	3	3	8	2	3	7		5	1	0
Santi Cazorla	p_8	0	10	11	13	5	7	5		8	7
Mustafi	p_9	6	13	1	1	2	2	0	15		13
Bellerin	p_{10}	1	0	2	1	4	5	0	2	11	
Coquelin	p_{11}	0	3	5	3	1	4	3	9	2	3

(clique) case. Each tie can be either present or absent; for each subgraph there are $2^{k(k-1)}$ possible states considering the exact identity of the individuals in the whole set.

Focusing on the structure elapsing between individuals, there are some recurrent states where the relational structure (i.e., in which way agents are linked between each other) is invariant under permutations of labels attached to the agents. Such categories are called *isomorphism classes*. The number of possible isomorphism classes of a k -subgraph depends on the number of elements belonging in each subgroup, in general this value can be exceedingly high even with modest subgroup dimensions, (Holland and Leinhardt 1976). For instance, when $k = 2, 3$ the number of isomorphism classes is 3 and 16, respectively, rapidly increasing even with $k = 4$ (218 classes).

In a purely theoretical point of view, both dyads and triads are two special cases of k -subgraphs. Figure 2 presents all the possible isomorphism classes derived from triad census, from the so called “003”, which involves no relationship between vertices, to the full connected triad “300”, presenting six edges. A summary of triads’ labels can be found in Figure 2.

Main differences between dyads and triads have been surveyed in Krackhardt (1999) and he stated that the individuality of agents is preserved when the analysis is focused on dyads, helping to highlight the direction of interactions between couples of agents. The census of all possible triads has been introduced by Davis and Leinhardt (1967), while Wasserman and Faust (1994) and Faust (2010) help to develop the method in order to retrieve information about network structural properties regarding the level (or quality) of transitivity nodes between individuals (see also Lorrain and White 1971; Luczkovich et al. 2003; Cugmas, Ferligoj, and Žiberna 2017, for further details).

In order to analyse the probabilistic properties of triad census, we use the notation introduced in Wasserman and Faust (1994) starting from the general k -subgraph theory. Then we apply authors’ results shown in Theorem 14.1 and 14.2 to the triadic case ($k = 3$), in order to obtain moments of triad census. We remark that these quantities are based on the theoretical distribution related to the random structure.

Given a network having n agents, let us suppose to label each node with a number equal to $1, 2, \dots, n$. We denote two distinct isomorphism classes for the k -subgraph η and ν , defined as two k -subgraphs having at least one vertex or edge that they do not share (while all other vertices and edges can overlap). We refer to generic

elements of the two isomorphism classes as $K = K_{i_1, \dots, i_k} \in \eta$ and $L = L_{i'_1, \dots, i'_k} \in \nu$, where the subscripts are related to the labels attached to each vertex of the subgraph. Therefore, we define the following probabilities:

$$p_K(\eta) = P(K \in \eta) \quad (1)$$

$$p_{K,L}(\eta, \nu) = P(K \in \eta \cap L \in \nu) \quad (2)$$

representing the probability that any single subgraph K or any two subgraphs K, L belong to classes η and ν , respectively. Since the above probabilities may vary for each configuration, it would be necessary to average the probability of each isomorphism class over all possible K and L structures available in the k -subgraph, as follows:

$$\bar{p}(\eta) = \frac{1}{\binom{n}{k}} \sum_K p_K(\eta) \quad (3)$$

$$\bar{p}_h(\eta, \nu) = \frac{1}{\binom{n}{k} \binom{n-k}{k-j} \binom{k}{j}} \sum_{|K \cap L|=h} p_{K,L}(\eta, \nu), \quad (4)$$

where $0 \leq h \leq k - 1$.

The subgraph census’ random variable H_η is then defined as the number of k -subgraphs belonging to the isomorphism class η . Using the theorem introduced Holland and Leinhardt (1976), it is possible to define the moments of k -subgraph census random variables H_η for each admissible isomorphism classes that can be retrieved in the k -subgraph. Hence, such moments rely on a given probability structure.

3 Testing triadic patterns

3.1 Testing against random structure

Several possible random distributions can be used to compute moments for each isomorphism (see Holland and Leinhardt 1976; Faust 2007, for further details). Among others, the so called U|MAN (Uniform given Mutual Asymmetric and Null arcs) is one of the most relevant structures in the literature, and will be used as reference throughout the

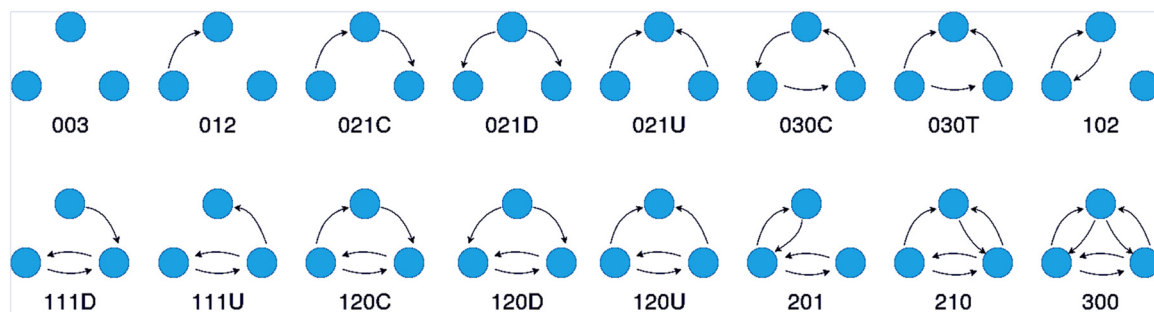


Figure 2: List of isomorphism classes belonging to the triadic subgraph. Each label consists in three digits representing the number of mutual ($\leftrightarrow$), asymmetric ($\rightarrow$), and null links connecting two units. Some classes with the same MAN distribution have a letter representing the additional structural differentiation: C = cycle, T = transitive, U = up and D = down. Plot adapted from Faust (2008).

rest of the paper. U|MAN random digraph is the probability distribution on the set of all digraphs with n agents considering fixed and discrete values of mutual, asymmetric and null arcs present in the network. The motivation to use the U|MAN distribution can be found in the possibility to construct, imposing a set of assumptions, an asymptotic statistical test for each of the 16 isomorphism classes and, more importantly, a comprehensive asymptotic test involving all types of triads.

Thus, it is possible to define a random network structure starting from the number of mutual, asymmetric and null arcs in the network comparing the observed triads against those which are uniformly distributed among the agents. In order to make this comparison, our purpose is to formally test if the probabilistic structure of a team passing network is closer or not to the one defined in the conditional uniform random case.

Since it holds a direct connection between averaged and single probabilities shown in Equations (1) and (2), the use of U|MAN distribution allows to simplify results of Equations (3) and (4) in the following way:

$$\begin{aligned}\bar{p}(\eta) &= p(\nu) \\ \bar{p}_{K,L}(\eta, \nu) &= p_h(\eta, \nu), \quad h = |K \cap L|\end{aligned}$$

where

$$\begin{aligned}p(\eta) &= P(K \in \eta) \\ p_h(\eta, \nu) &= P(K \in \eta \cap L \in \nu), \quad h = |K \cap L|\end{aligned}$$

and $0 \leq h \leq k-1$. The first equality is a consequence of the invariance under permutations property of U|MAN distribution; to summarize, changing the node' labels of a triad does not change the probability of belonging to an isomorphism class. On the same basis, the second equation indicates that the probabilities $p_{K,L}(\eta, \nu)$ depend only on η and ν and also on the number of nodes they share, i.e., $h = |K \cap L|$.

Moreover, it is possible to derive a correspondence between the joint probabilities of two triads having either zero or one vertex in common, such as:

$$p_0(\eta, \nu) = p_1(\eta, \nu).$$

Basing on the above statements, it is possible to apply the results shown in Holland and Leinhardt (1976) under the condition that data generating process of links in a network is characterized by a U|MAN distribution. As mentioned, the total number of isomorphism classes of a triad census is 16 (as in Figure 2) and it is possible to define the triad census vector $\mathbf{T} = (T_1, \dots, T_{16})$ from a random directed graph based on the U|MAN distribution. Expectation, variance, and covariance for each isomorphism class are directly

retrieved combining the results in Holland and Leinhardt (1976) and are depicted in the Appendix A. Regarding the asymptotic distribution of vector $\mathbf{T}$, it is worth to notice that it has an approximate multivariate normal distribution with an increasing number of vertices included in the graph (Wasserman and Faust 1994). Starting from results presented in the Appendix A, Equations (9)–(11), it is possible to compute the τ_η statistics for each isomorphism class

$$\tau_\eta = \frac{T_\eta - E(T_\eta)}{\sqrt{\text{Var}(T_\eta)}}. \quad (5)$$

For each class η , τ_η represents the standardized departures of observed triads from the theoretical random distribution, allowing to formally test the presence of patterns that deviate from what is expected in the randomly distributed case. Thus, a test statistic denoted as $\tau_{\max}^2$ can be used to build an omnibus test of randomness:

$$\tau_{\max}^2 = (\mathbf{T} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{T} - \boldsymbol{\mu}) \quad (6)$$

where $\boldsymbol{\mu}$ is the vector of expectations and $\boldsymbol{\Sigma}$ the variance–covariance matrix computed using the U|MAN distribution. $\boldsymbol{\Sigma}^{-1}$ is the pseudo-inverse of the variance–covariance matrix and can be obtained as terms of eigenvalue decomposition $\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Gamma} \boldsymbol{D}^{-1} \boldsymbol{\Gamma}'$, where $\boldsymbol{\Gamma}$ is the matrix of eigenvectors of $\boldsymbol{\Sigma}$ and the values λ_i of the inverse diagonal matrix $\boldsymbol{D}^{-1} = \text{diag}(1/\lambda_1, \dots, 1/\lambda_p, 0, \dots, 0)$ correspond to the non-null eigenvalues of $\boldsymbol{\Sigma}$, for $i = 1, \dots, p$ and $p = \text{rank}(\boldsymbol{\Sigma}) \leq 16$. The above statistic can be used as test for detecting the possible deviation against a random network structure in the data.

Holland and Leinhardt (1978) prove that, given a random reference distribution and if the vector $\mathbf{T}$ is approximately Normal, $\tau_{\max}^2$ asymptotically follows a χ^2 distribution with degrees of freedom equal to the rank of $\boldsymbol{\Sigma}$. Therefore, we can test if data exhibit a random structure compared to the alternative hypothesis of strong deviations against randomness.

3.2 Testing specific team strategies

The approach introduced in Holland and Leinhardt (1978) lacks of flexibility: it is not straightforward to manipulate the null hypothesis in order to test for deviation from different random structures or, more importantly, to test for significant changes in a subset of triad counts. Moreover, the $\tau_{\max}^2$ test statistic may be more conservative in finite samples.

A possible workaround could be the use of univariate tests for each isomorphism class, but this alternative generally implies the fulfilment of strict assumptions (such

as independence between the triadic configurations) that are not satisfied in this context. Then, given the multivariate nature of triadic configurations, we are interested in defining a set of multivariate hypotheses allowing to formally test a set of factors composed by non-independent counts.

Testing procedures of count data have been applied in other fields, such as the analysis of contingency tables or ratings data obtained from ordered categorical responses showing the presence of variability between samples (Ennis and Bi 1999), but mainly related with biology and genetics, i.e., olfactometer data (Ricard and Davison 2007) or DNA sequencing data (Wu et al. 2017). A comprehensive analysis of discrete multivariate distributions can be found in Johnson, Kotz, and Balakrishnan (1997) for a.

Despite the presence of interest in literature regarding this topic, to the best of our knowledge there are no applications to triad census analysis. In this context, due to the structure induced by isomorphism classes, the Dirichlet-Multinomial (hereafter DM) distribution (introduced in Mosimann 1962) suits properly the features belonging to triad census data. Throughout the rest of the paper we will refer to Equation (13) (see Appendix B) and then we will denote that an m -variate random variable follows a DM distribution as $\mathbf{X} \sim \text{DM}_m(\boldsymbol{\pi}, \theta)$. The main information regarding DM distribution useful for the purpose of the present work is depicted in the Appendix B.

Beta-Binomial distribution is a special case of the DM distribution when the number of modalities reduces to $m = 2$. It can also be shown that it approaches the Multinomial distribution as θ approaches to zero. Parameter estimation can be carried out through Maximum Likelihood or Method of Moments as described in Minka (2000) and Brier (1980).

Our proposal is to adopt DM family of distributions to the problem of formally testing for the presence of “structural equivalence” in a set of triad census data. The aim of the test is to determine if a sample belongs to a previously specified distribution, given that the data generating process is from a DM family.

Initially, Brier (1980) focused their effort on the study of fitting and testing of general log-linear models in contingency tables following DM distribution, then the problem of testing vector of proportions generated from different families of processes has been surveyed (Koehler and Wilson 1986), deriving general goodness-of-fit statistics based on generalized Wald statistic, also computing asymptotic results.

More recently, a goodness-of-fit statistic for testing the fit of a model based on DM family is also obtained in Johnson, Kotz, and Balakrishnan (1997): given that data have

been generated by a DM distribution, a possible approach could be to compare the estimated frequencies with a reference distribution defined in null hypothesis by imposing theoretical frequencies $\boldsymbol{\pi}_0$. Nowadays, with the One-Sample DM test (OSDM) for grouped count data, La Rosa et al. (2012) extended this testing procedure to formally test for a set of g samples belonging to the same population measured on different subjects.

Consider a set $\{\mathbf{x}_i\}_{i=1,\dots,g}$ of g samples, each one consisting of m counts of distinct modalities where the count of modality i corresponding to the j th sample is denoted as $x_{i,j}$. Denote $x_{0,\bullet} = \sum_{i=1}^m \sum_{j=1}^g x_{i,j}$ as the sum of all frequencies over all samples, while $x_{0,j} = \sum_{i=1}^m x_{i,j}$ stands for the total of the j th sample. Then, to formally tests the hypothesis $H_0: \boldsymbol{\pi} = \boldsymbol{\pi}_0$ versus the alternative $H_1: \boldsymbol{\pi} \neq \boldsymbol{\pi}_0$ the following test statistic is presented

$$X_{\text{obs}} = (\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0)^\top \Sigma^+ (\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \quad (7)$$

where the operator $(\cdot)^+$ is the Moore–Penrose generalized inverse and the matrix Σ corresponds to

$$\frac{1}{x_{0,\bullet}^2} \left[\hat{\theta} \left(\sum_{j=1}^g x_{0,j}^2 - x_{0,\bullet} \right) + x_{0,\bullet} \right] (\text{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top) \quad (8)$$

with $(\hat{\boldsymbol{\pi}}, \hat{\theta})$ unbiased estimators of distribution parameters. The test statistic X_{obs} asymptotically converges to a χ^2 distribution with degrees of freedom equal to the rank of the matrix $(\text{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top)^+$.

In addition, due to the aggregation property (see, e.g., Johnson, Kotz, and Balakrishnan 1997, and references therein), the DM family allows for a flexible re-parametrization of the probability vectors: the joint distribution over sums of disjoint subsets of modality counts is also DM, i.e., if $(x_1, \dots, x_m) \sim \text{DM}_m(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_m, \theta)$ then, $(x_1, \dots, x_s, x_0 - \sum_{i=s+1}^m x_i) \sim \text{DM}_{s+1}(\boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_s, 1 - \sum_{i=s+1}^m \pi_i, \theta)$. Hence, it is possible to “collapse” a subset of isomorphism classes into one containing the sum of their counts, e.g., the ones having low frequencies or structural zeroes.

We redesign the testing procedure in Equation (7) to triad census analysis to unveil if a triadic pattern, computed from a group of networks sharing similar features, has been generated from a reference DM distribution. To this end, we propose two sets of null hypotheses. Firstly, reference team strategies (based on the triadic configuration) are set to identify if and how teams follow a given strategy. Secondly, a more direct comparison between couple of teams can be performed to detect possible similarities.

Although a small-scale simulation of size has been performed considering either few m modalities and small total

counts $x_{0,\bullet}$ (see La Rosa et al. 2012), there is no empirical evidence in literature of test statistic validity when the number of g samples is small, as it happens in our empirical application. For this reason, in order to give a further support to our testing strategy, we carried out a small scale Monte Carlo exercise computing the empirical size of the DM test with small sample sizes. All the details are given in Appendix.

4 Empirical application

In this Section we summarize the main results of data analysis pertaining to Football Data. Our data consist of a set of team passing distributions coming from the Group stage of 32 UEFA Champions League (UCL) for three consecutive Seasons: 2016–2017, 2017–2018 and 2018–2019. Data were collected through the official UEFA website (www.uefa.com) and subsequently processed to include 96 matches and 192 passing networks for each season, for a total of 288 matches and 576 passing networks. Fifty-six teams were involved; among them, 26 teams (46.4 %) played in nearly two editions of the tournament, while 14 teams (25 %) were qualified in the competition in all editions. England, Spain, Germany and Italy were the most represented federations, followed by France and Portugal. Table 2 depicts the overall descriptive statistics of passes in the three considered UCL Seasons for the teams. The mean of passes for each team in a match was about 386 passes (median equal to 370). The variability around the mean does not indicate substantial changes among the three UCL Seasons, even if the range can be very large.

The total number of passes contributes to the perceived competitiveness of a team. In Figure 3 we include only the best and worst five teams in the first phase of the three considered UCL editions, highlighting the best and worst performances of their first lineups. In terms of passes, all of

the top-five-ranked teams advanced to the knockout phase, with the exception of Naples in 2018–2019, although that team did achieve the same number of points as the second-best team in the group (Liverpool). Barcelona (Spain) was the only team that was always in the top five, while Borussia Dortmund, Real Madrid, Bayern München and Manchester City placed in higher positions of the ranking twice over three times.

Furthermore, four-fifths of the “worst” five teams in terms of completed passes failed to pass the Group Stage in each edition. Only Leicester City (2016–2017), Basel (2017–2018) and Schalke 04 (2018–2019) managed to advance despite their poor performance. Among them, only Leicester City passed through the round of 16 to get to the round of 8.

In the following, triad census methods are applied to our networks, including the formal tests of Section 3. The main results were developed through R software using *igraph* (Csardi and Nepusz 2006) and *HMP* (La Rosa et al. 2019).

4.1 Descriptive analysis

Network structure can be relevant in this kind of analysis, showing evidence of having relationship with the number of passes. For example, considering all networks, a positive and statistically significant large correlation can be found between the number of passes and the mutual dyads (0.77, p -value < 0.001). In addition, the asymmetric and null dyads were negatively correlated with the passes (−0.67 and −0.45 respectively), again with statistical significance (p -value < 0.001 for both variables).

Regarding the dyadic distribution, the football teams expressed a median of 37 mutual relationships per match, while unilateral links varied between 10 and 15 in 50 % of the teams. There was an average of 6 unconnected dyads

Table 2: Descriptive statistics of total number of passes in football UCL data.

	Statistic	2016–2017	2017–2018	2018–2019	Overall
Total passes	Mean	386.25	388.34	382.19	385.60
	SD	136.93	134.31	131.30	133.99
	min	121.00	126.00	114.00	114.00
	Q1	285.75	278.50	278.75	281.00
	Median	373.00	377.00	362.00	370.50
	Q3	462.25	475.50	471.00	467.75
	Max	876.00	743.00	921.00	921.00

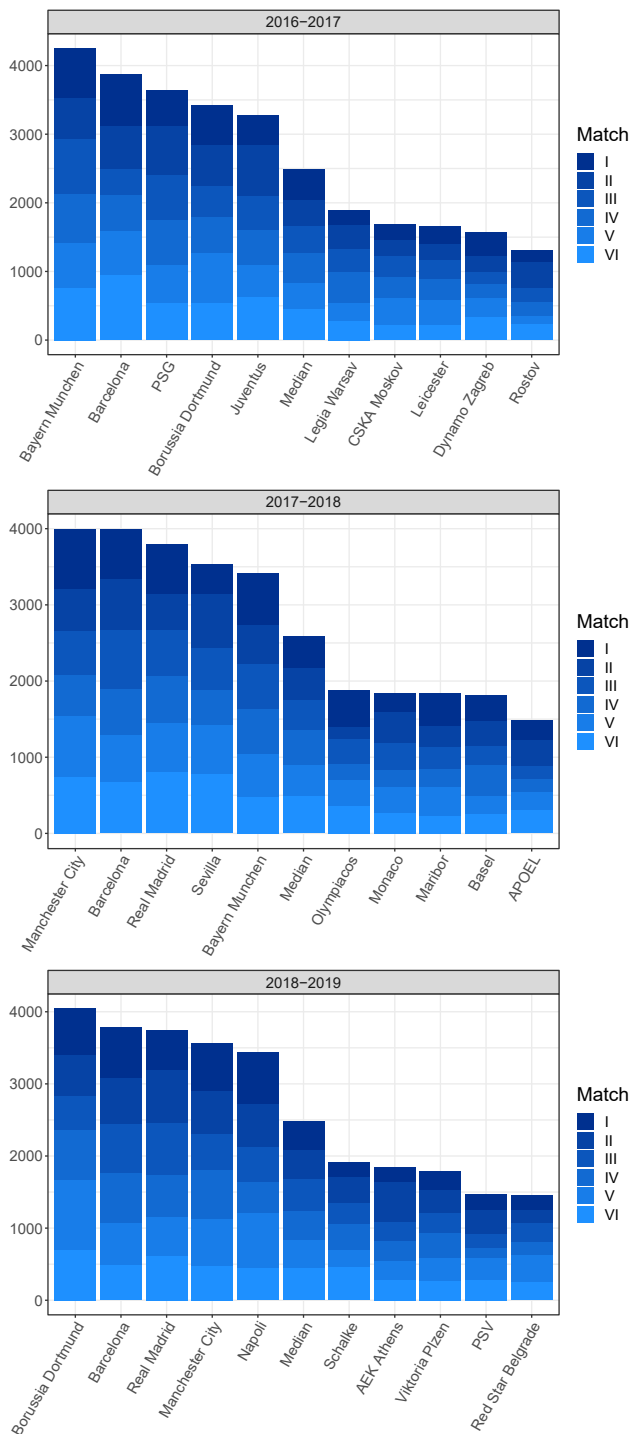


Figure 3: List of best and worst five teams (along with median values) ranked by the number of completed passes in each season.

for each team per match, varying between a minimum of 0 and a maximum of 19, as well as presenting higher variability (47 %) among the MAN. Differences between the three seasons seemed to be negligible for the three counts. We

must remark that the sum of MAN for football teams ($n = 11$ players) is constrained to 55, i.e., $(n(n-1)/2)$.

Furthermore, generalizing to the triad census, we recognized specific triadic patterns in football passes. Table 4 shows the distribution of triads for passing networks of all teams involved in the group stage of the UCL 2016–2017 to 2018–2019 seasons ($n = 576$).

We highlighted noticeable differences in terms of mean and variability between 16 isomorphism classes. In particular, the most frequent triads were as follows: “300” (fully connected), which also presented the highest variability; “210”; “201” and “111D”. Additionally, classes with lower ties, involving sparse connections, seemed to be recurrent structures in these types of networks. For example, if we consider classes with “0” as first number (corresponding to all the isomorphism classes having zero mutual edges), we can also observe median values greater than zero and a certain amount of variability, expressed in terms of standard deviation. Moreover, several isomorphism classes appeared less frequently in football games, such as the types “030C” and “030T”, involving three directed links.

These findings revealed uncommon types of triadic relations in passing networks: type “030C” refers to cyclic ties, i.e., $A \rightarrow B \rightarrow C \rightarrow A$, while “030T” represents transitive triads of the form $A \rightarrow B \rightarrow C, A \rightarrow C$. To summarize, such classes mainly concern transitivity structure and share the presence of three directed links (arrows). The reasons behind these low counts remain somewhat unclear, but could be linked to the direction of the ball (clearly related to the main aim of the game) and positions on the pitch (initial formation). Another issue can be represented by the possibility of “structural” zero issue (Block 2015). Some triad counts can be constrained to zero in particular conditions (i.e., the “003” case if all players nearly make a pass to each individual other player). In general, we remark that, similarly to the dyad census case, the total number of triads for each team ($n = 11$ players) in each match was constrained to 165, i.e., $\binom{n}{3}$.

Figure 4 shows the distribution of triad counts for the three UCL Seasons, confirming some considerations mentioned above.

In Figure 5, the first 10 teams are plotted according to their average value for all 16 isomorphism classes. We immediately noticed that skilful teams such as Real Madrid, Barcelona and Manchester City presented an average of “300” counts, which is always greater than 70. On the other hand, teams such as Anderlecht, Dinamo Zagreb and Leicester were more likely to present counts of disconnected or partially non-connected triads (such as “003” and “012”).

Table 3: Descriptive statistics of MAN in football UCL data.

Edge type		2016–2017	2017–2018	2018–2019	Overall
Mutual	Mean	36.61	36.16	36.48	36.42
	SD	5.32	5.41	5.10	5.27.00
	Min	17.00	19.00	21.00	17.00
	Q1	33.00	33.00	34.00	33.00
	Median	37.00	37.00	36	37.00
	Q3	40	40.00	40.00	40.00
	Max	46	49.00	47.00	49.00
Asymmetric	Mean	12.21	12.51	12.57	12.43
	SD	4.13	3.91	4.10	4.04
	Min	3.00	4.00	3.00	3.00
	Q1	9.00	10.00	9.00	10.00
	Median	12.00	12.00	12.00	12.00
	Q3	15.00	15.00	15.00	15.00
	Max	26.00	23.00	25.00	26.00
Null	Mean	6.17	6.33	5.94	6.15
	SD	2.94	3.04	2.75	2.91
	Min	0.00	1.00	0.00	0.00
	Q1	4.00	4.00	4.00	4.00
	Median	6.00	6.00	6.00	6.00
	Q3	8.00	8.00	8.00	8.00
	Max	17.00	19.00	15.00	19.00

Table 4: Summary statistics of triad census.

Triad	Mean	SD	Min	Q1	Median	Q3	Max
003	0.33	0.93	0	0	0	0	10
012	1.87	2.60	0	0	1	3	23
021C	1.70	2.03	0	0	1	2	13
021D	0.84	1.22	0	0	0	1	9
021U	0.89	1.22	0	0	1	1	7
030C	0.48	0.85	0	0	0	1	5
030T	1.90	2.46	0	0	1	3	20
102	5.43	4.82	0	2	4	7	29
111D	9.70	4.42	0	7	10	12	25
111U	8.55	4.71	0	5	8	11	24
120C	7.36	4.63	0	4	7	10	22
120D	5.60	3.99	0	3	5	8	27
120U	4.82	3.43	0	2	4	7	21
201	18.11	7.49	0	13	18	23	43
210	42.17	9.44	16	36	43	48	70
300	55.24	20.28	6	41	54	69	118

In keeping with this, triadic information could be summarized by factor analysis. We followed the approach of Faust (2006) to investigate the “resemblance” of triad census in empirical networks, although several approaches may be conducted to make dimensionality reduction (see Greenacre 1984, for a comprehensive overview). In practice,

we applied the Correspondence Analysis (CA) to the original data, where the 16 isomorphism classes can be viewed as the modalities of a single variable by looking at data as a cross-tabulation between two categorical variables. Each row of the input matrix represents a team in a match that occurred during the Group Stage of the UCL, while the counts of all the isomorphism classes are arranged in columns.

Considering all networks, the first two dimensions of the CA explain the more than 60 % “triadic” variability, with 38.4 % for the first dimension and 22.9 % for the second. The third dimension also explains the 6.3 % “triadic” variability.

The scree plot of Figure 6 suggests the presence of at least three factors (67.6 % of the overall variability) according to the conventional Kaiser criterion (Kaiser 1960). According to the above results, we focused on these three dimensions. Furthermore, Figure 7 shows both individuals (passing networks of teams) and isomorphism classes on the factor map in first two dimensions. The first dimension is mainly characterized by the presence or absence for connections, with opposing classes such “012C”, “021D” and “021U” against the fully connected type, i.e., “300”. Hence, the second dimension is mainly driven by differences between “210”, i.e., including two transitive connections and two disconnected vertices, and “210”, presenting five links over six.

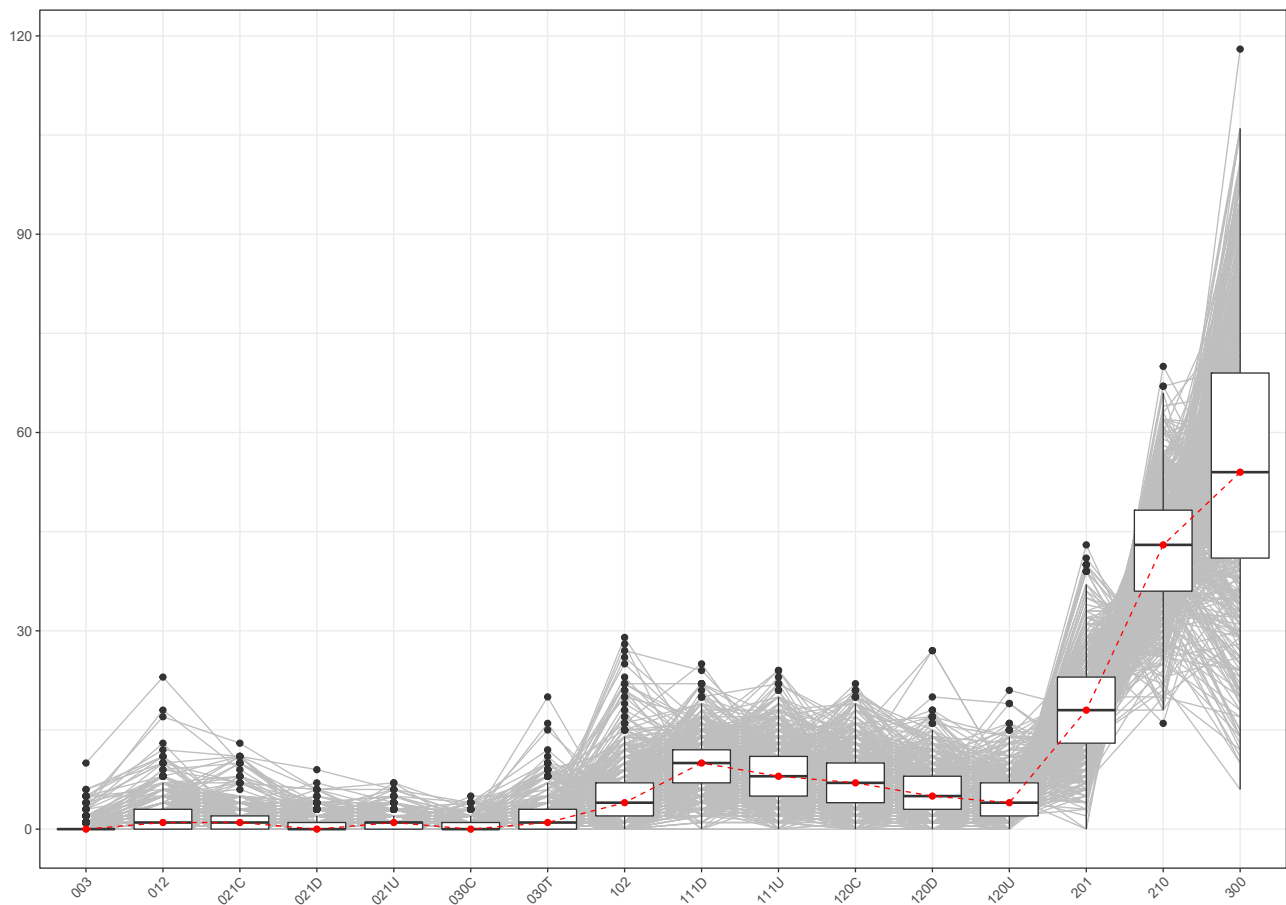


Figure 4: Distribution of isomorphism classes in Group Stage of Uefa Champions League ($n = 576$).

Then, in Figure 8, we can observe the contribution of all 16 isomorphism classes for the first three CA dimensions in terms of squared cosines (or squared correlations). These quantities directly express the acceptability of the triad projections along the three dimensions; that is, these quantities can be used to measure the quality of the representation of each isomorphism class on the factor map. For the first dimension, the “300” class is the best represented in terms of squared cosine, reaching a high value of 91 %, followed by “012C”, “012” and “111U” with values higher than 45 %. In general, only “201” and “210” appear to be almost orthogonal with respect to the first factor. Regarding the second dimension, satisfyingly represented classes in terms of squared cosine were “102” (including only a single mutual link between three agents), as well as “120” and “210”, presenting two transitive links. Interestingly, the three classes “012D”, “012U” and “012C” are not correlated with the second factor. Only classes “003” and “201” appeared to be correlated with the third dimension.

We also graphically identify the isomorphism classes that presents the higher contribution to the first three CA dimensions (Figure 9). Classes “300”, “120”, “012” and “210” are greater than the empirical threshold of $1/(m-1)$, while “030T” and “003” are almost equal to this value. Other triads appear redundant to explain the variability of the football passing network.

Finally, starting from these results, we applied k -means clustering in order to establish a set of motivated profiles, defining a proper null hypothesis for the OSDM test. This allowed us to identify a set of triadic profiles that are related to different triadic behaviours in football. We chose $\kappa = 4$ clusters based on the total within-cluster sum of squares.

The four resulting centroids are plotted on the factor map as supplementary points in Figure 7, and their reconstructed triads are found in Table 5 and compared with the general median of the overall teams.

Table 5, shows that the first centroid T_1 presents a number of “102” and “201” greater than the general medians, while the second one (T_2) presents lower “201” and greater

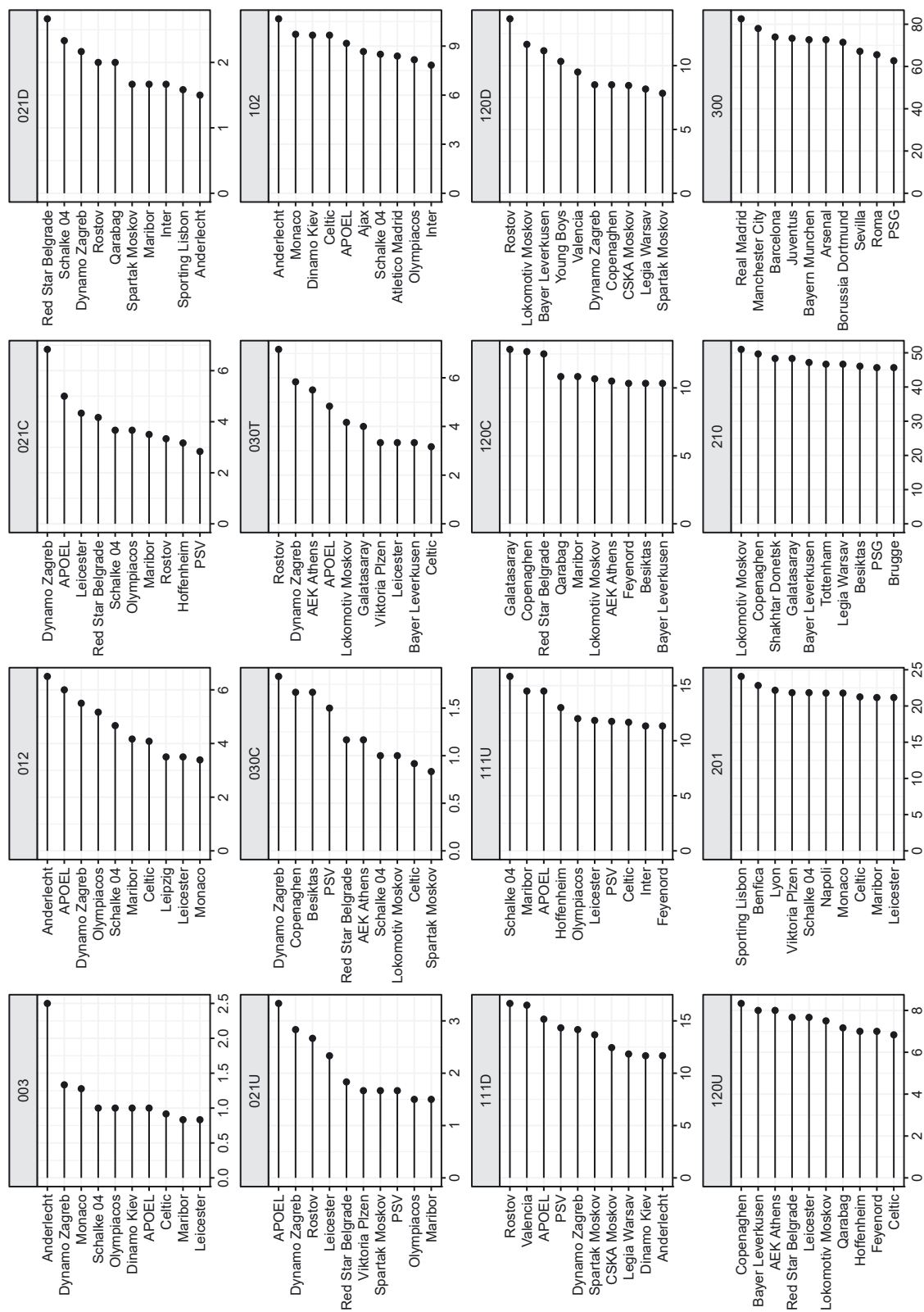


Figure 5: First 10 European Teams for all isomorphism classes (average values).

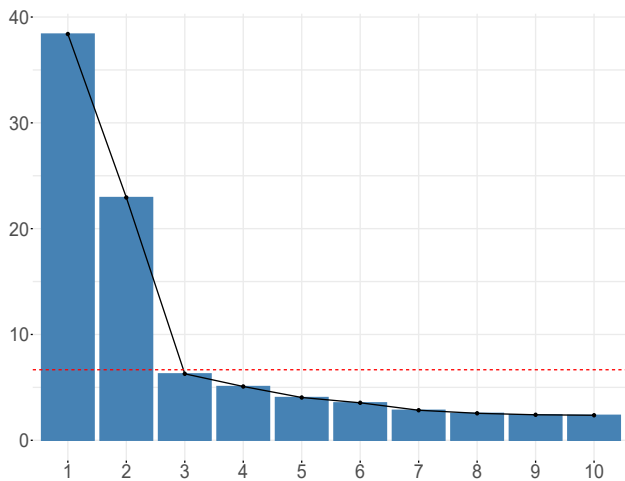


Figure 6: Scree plot of the CA on triad census. Red dashed line indicates the reference value $1/(m-1)$, where m is the number of modalities.

“210” and “120” (D, U, C) types. The centroid denoted as T_3 shows the highest counts of fully connected triads, i.e., “300” (exceeding the general median of 22 units), while on the contrary, centroid T_4 exhibits the lower value of this class

and the higher counts of triads with a low number of links, i.e., “012” and “102”.

4.2 Testing the presence of a strategy

After performing a statistical comparison of triad census, we empirically evaluated differences between the observed triads of the football teams against their conditional random reference, i.e., the U|MAN, following the methodology presented in Section 3.1.

Figure 10 graphically shows the distribution over three UCL Seasons of τ_u statistics as defined in Section 3.1. As expected, such statistics did not seem to show substantial changes among seasons both in terms of location (mean or median) and scale (variability expressed through interquartile range), although there were some exceptions (see, for example, type “030C”). Moreover, most connected classes, i.e., “201”, “210” and “300” as well as “102” and “111D”, present evidence against the theoretical values of the random graph scenario, i.e., the null hypothesis discussed in Section 3.1 (here represented by a dashed red line) more often than others.

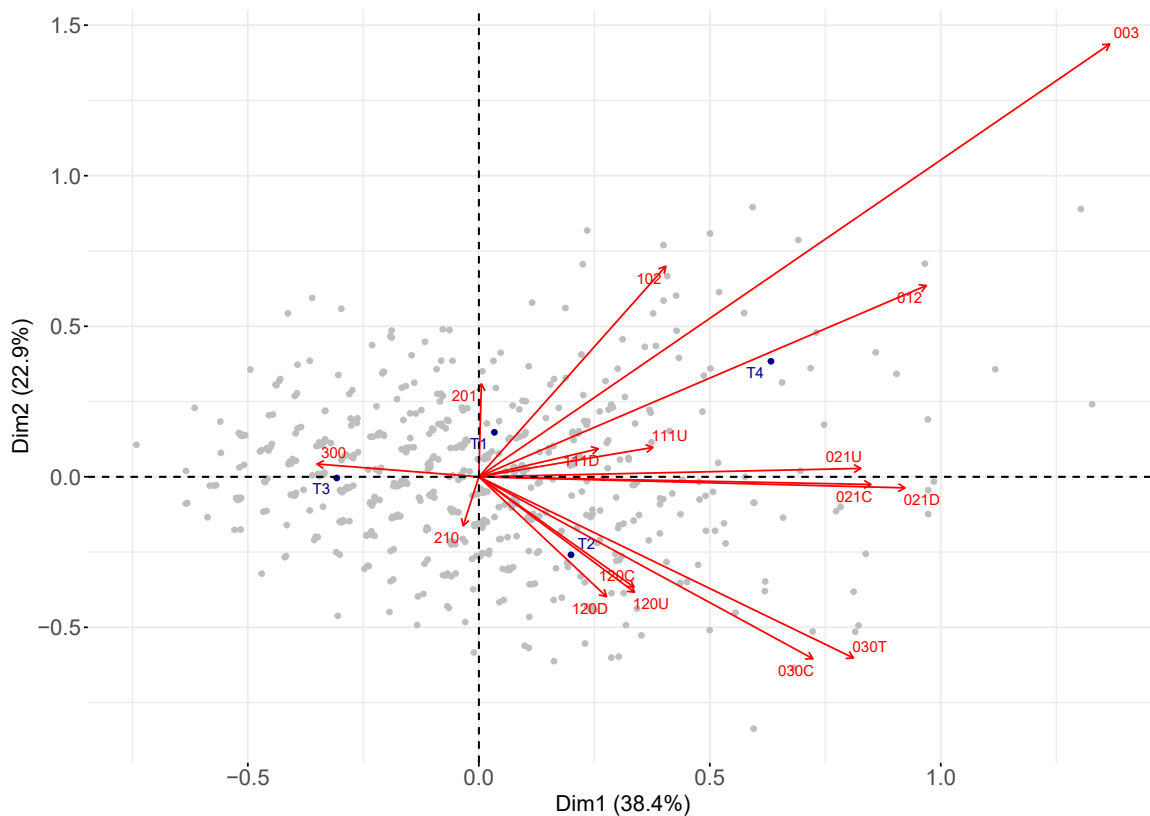


Figure 7: Biplot of first two CA factors. Red arrows are isomorphism classes and grey points are the networks (matches). Blue supplementary points represent coordinates of four k -means centroids explained in the text.

	003	012	102	021D	021U	021C	111D	111U	030T	030C	201	120D	120U	120C	210	300
Dim 1	0.24	0.49	0.21	0.41	0.36	0.51	0.32	0.47	0.39	0.17	0	0.15	0.23	0.29	0.02	0.91
Dim 2	0.26	0.21	0.62	0	0	0	0.04	0.03	0.22	0.12	0.56	0.31	0.29	0.34	0.53	0.01
Dim 3	0.24	0.14	0	0	0.01	0.05	0.01	0.15	0	0	0.27	0	0.03	0.01	0	0.04

Figure 8: Heatmap of squared cosines related with triads in the CA dimensions.

In order to carry out the formal test introduced in Section 3.1, we computed $\tau_{\max}^2$ statistics for all available UCL matches. Table 6 contains the percentage of p -values associated with $\tau_{\max}^2$, which are lower than conventional nominal levels, i.e., 0.1, 0.05 and 0.01, even divided by the three Seasons (576 matches). Since we are considering multiple testing, we also included the rejection frequencies adjusted using the Bonferroni correction (i.e., $p = \alpha/G$). The proportion of p -values lower than nominal levels can vary from 24 % (when the nominal level is 0.01) to 53 % with a greater I error type, and it is noticeable that first Season (2016–2017) presents the highest number of rejections. Using the Bonferroni correction for multiple testing, we observed lower rejection frequencies, and the null hypothesis of U|MAN structure can be rejected for one-tenth of the networks.

Despite these unexciting results, we should remark that (a) a test based on $\tau_{\max}^2$ statistic may be more conservative in finite samples, mainly due to the well-known asymptotic approximation issue, and (b) the (conditional) distribution of $\tau_{\max}^2$ under the null hypothesis of triadic randomness strictly depends on the linear dependence occurring in the

triad census matrix (i.e., its rank). For example, in our football data, we found a mode of 13 for the ranking of the triad census matrix among all networks. As discussed in Section 3.1, this number corresponds to the degrees of freedom in the $\tau_{\max}^2$ statistic under the null hypothesis of a random graph.

Ultimately, we can conclude that more than 40 % of the football teams involved in a match showed strong empirical evidence in terms of triadic behaviour against their random graph counterparts based on U|MAN.

Table 7 contains the rejection frequencies, at a 5 % nominal level, for European teams during the three Seasons. PSG rejected all the times in Season 2017–2018, followed by Real Madrid, Porto and Rostov also rejecting the randomness five times out of six in the previous Season. Considering all Seasons, Atletico Madrid rejected the null hypothesis two times out of three (12 matches), followed by PSG (9 matches over 18). Surprisingly, team such as Borussia Dortmund rejects the null for only one match over 18.

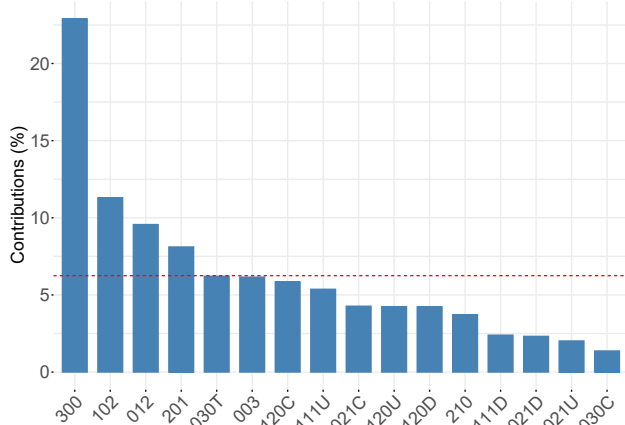


Figure 9: Contributions of isomorphism classes to the CA dimensions. Red dashed line is the reference value $1/15$.

Table 5: Reconstructed Triad census of k -means based centroids.

Class	T_1	T_2	T_3	T_4	Median
003	0	0	0	2	0
012	2	2	1	9	1
102	8	3	3	16	4
021D	1	1	0	2	0
021U	1	1	0	3	1
021C	2	2	1	5	1
111D	11	11	7	14	10
111U	11	10	5	14	8
030T	1	4	1	3	1
030C	0	1	0	1	0
201	24	13	16	22	18
120D	4	9	4	5	5
120U	4	8	3	5	4
120C	6	12	5	8	7
210	39	48	43	30	43
300	51	40	76	26	54

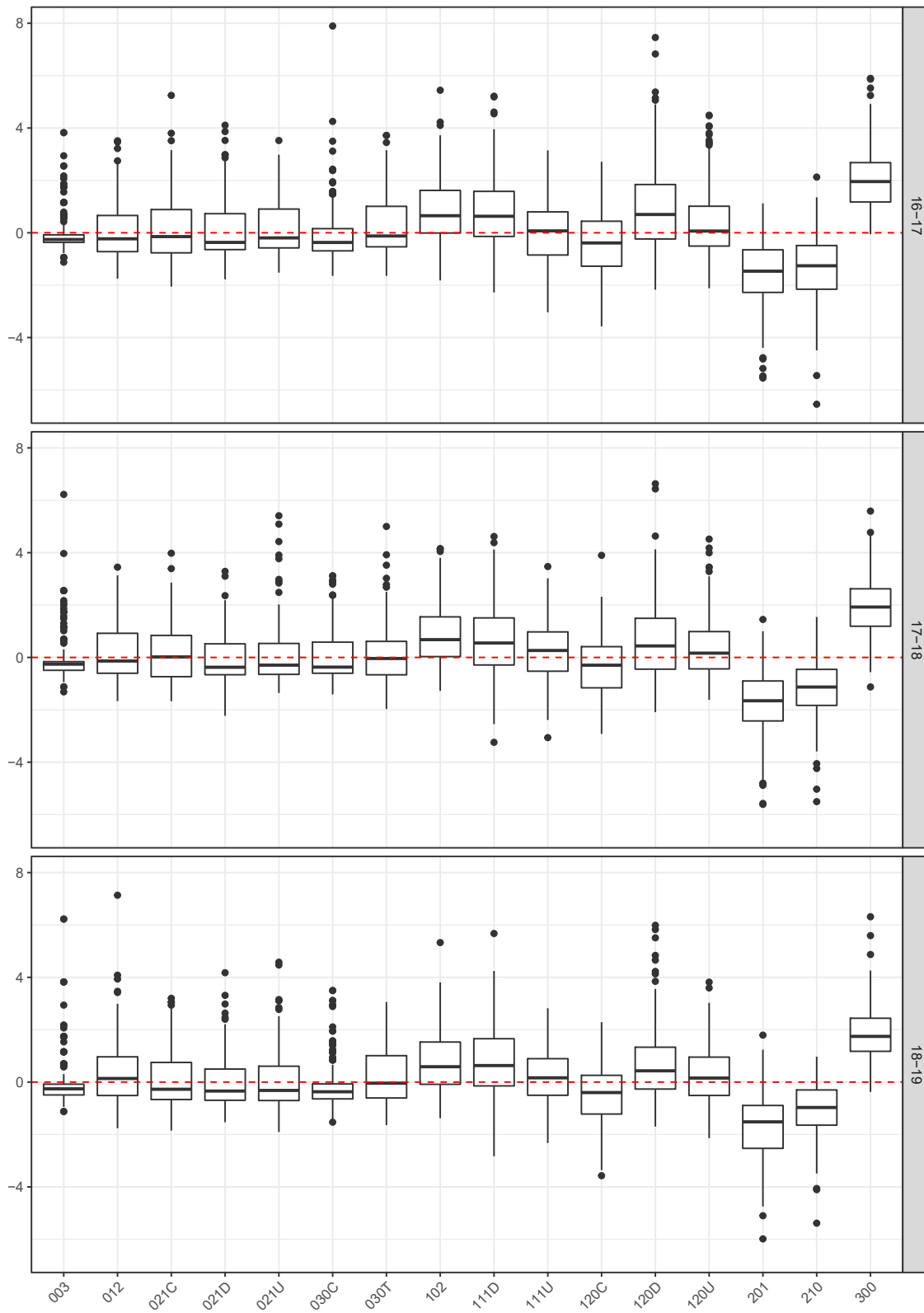


Figure 10: Boxplot of τ_u statistics, showing discrepancies between triad frequencies and expected frequencies under U|MAN distribution per Season.

Table 6: Percentages of rejection at a fixed nominal levels (Conventional – Conv) and using Bonferroni's (Bonf) correction for the $\tau_{\max}^2$ statistics.

Season	$p < 0.1$		$p < 0.05$		$p < 0.01$	
	Conv	Bonf	Conv	Bonf	Conv	Bonf
2016–2017	53 %	11 %	44 %	10 %	31 %	9 %
2017–2018	43 %	11 %	35 %	8 %	24 %	7 %
2018–2019	43 %	8 %	35 %	7 %	24 %	6 %
2016–2019	46 %	10 %	38 %	8 %	27 %	7 %

The reasons of those rejections can be significantly vary. In some cases, they are driven by high levels of isomorphism classes with low connection (i.e., “012” or “120”); in other cases, the teams are able to exhibit a large number of connected triads (i.e., types “300” or “210”), as well as

providing the expected values of their MAN distribution. If we consider the adjusted error rate to be equal to 0.05/192, we can observe that the test was rejected two times out of six (in each Season) for the following teams: Manchester City and Benfica (2016–2017), Real Madrid and Anderlecht (2017–2018), and Valencia (2018–2019).

4.3 Triad census and style of play

We further tested for the presence of a “structural equivalence” in terms of triadic relationships applying the OSDM test outlined in Section 2.2. First, the OSDM test was applied considering the six matches of each team in the three Seasons. Thus, we formalized four null hypotheses based upon the four cluster centroids reconstructed in Table 5 of Section 4.1.

Table 7: Number of matches in which each team rejects the $\tau_{\max}^2$ in each Season at 5 % significance level. Results based on Bonferroni's correction are depicted in brackets.

Season 2016–2017	$p < 0.05$	Season 2017–2018	$p < 0.05$	Season 2018–2019	$p < 0.05$
Porto	5 (1)	PSG	6 (1)	Atletico Madrid	4 (1)
Rostov	5 (1)	Real Madrid	5 (2)	Valencia	4 (2)
Arsenal	4 (1)	Atletico Madrid	4 (1)	Ajax	3 (1)
Atletico Madrid	4 (0)	Chelsea	4 (1)	Barcelona	3 (0)
Basel	4 (1)	Liverpool	4 (1)	Brugge	3 (0)
Bayer Leverkusen	4 (1)	Qarabag	4 (0)	CSKA Moskov	3 (1)
Bayern Munchen	4 (0)	Anderlecht	3 (2)	Hoffenheim	3 (0)
Celtic	4 (0)	Juventus	3 (0)	Inter	3 (0)
Manchester City	4 (2)	Napoli	3 (0)	Juventus	3 (0)
Brugge	3 (1)	Olympiacos	3 (0)	Lokomotiv Moskov	3 (1)
CSKA Moskov	3 (0)	Tottenham	3 (1)	Manchester United	3 (0)
Juventus	3 (1)	Barcelona	2 (0)	Monaco	3 (0)
Lyon	3 (0)	Besiktas	2 (1)	Red Star Belgrade	3 (1)
Monaco	3 (0)	Manchester City	2 (1)	Roma	3 (1)
Napoli	3 (0)	Porto	2 (1)	Bayern Munchen	2 (1)
PSV	3 (0)	Roma	2 (1)	Manchester City	2 (0)
Sevilla	3 (1)	Sevilla	2 (1)	Napoli	2 (0)
Tottenham	3 (1)	Spartak Moskov	2 (1)	PSV	2 (0)
Barcelona	2 (1)	Bayern Munchen	1 (0)	Real Madrid	2 (0)
Benfica	2 (2)	Benfica	1 (0)	Tottenham	2 (1)
Besiktas	2 (0)	Borussia Dortmund	1 (0)	Viktoria Plzen	2 (0)
Legia Warsaw	2 (1)	Celtic	1 (0)	Young Boys	2 (1)
Leicester	2 (0)	CSKA Moskov	1 (0)	AEK Athens	1 (0)
Ludogorets	2 (1)	Leipzig	1 (1)	Benfica	1 (0)
PSG	2 (1)	Manchester United	1 (0)	Liverpool	1 (0)
Sporting Lisbon	2 (0)	Maribor	1 (0)	Lyon	1 (0)
Borussia Gladbach	1 (0)	Monaco	1 (0)	Porto	1 (0)
Copenhagen	1 (0)	Shakhtar Donetsk	1 (0)	PSG	1 (0)
Dynamo Kiev	1 (1)	Sporting Lisbon	1 (0)	Schalke	1 (1)
Real Madrid	1 (1)	APOEL	0 (0)	Shakhtar Donetsk	1 (1)
Borussia Dortmund	0 (0)	Basel	0 (0)	Borussia Dortmund	0 (0)
Dynamo Zagreb	0 (0)	Feyenord	0 (0)	Galatasaray	0 (0)

In Tables 9–12 (depicted in Appendix D) we rank the teams according to their OSDM estimated statistics and their associated p -value in the three UCL Seasons. Regarding the first system, denoted as T_1 , it can be seen that the null hypothesis cannot be rejected for almost all Dutch and French teams (excluding the first Season of PSG) and the Spanish teams of Atletico Madrid and Seville. Similar to this strategy are most of the Italian teams, such as Naples, Rome (2017–2018) and Inter.

If we consider the null hypothesis constructed upon T_2 , few teams were ultimately reflected in this strategy. Notably, however, all Russian teams (CSKA Moscow, Spartak and Lokomotiv Moscow) were unable to reject the hypothesis at a 5 % nominal level throughout the three UCL seasons. These teams are followed by other “Eastern” Teams such as Qarabag, Galatasaray, AEK Athens and Warsaw.

As mentioned, the third cluster (T_3) contains teams that generally produce more closed triads between players. For all three years, we found that the OSDM could not be rejected for Manchester City, Juventus, Real Madrid, Bayern Munchen, Borussia Dortmund and Barcelona (except the preceding Season). Finally, almost all teams rejected the null hypothesis T_4 based on the four centroids.

Another possibility is given by the use of “reference” teams as null hypothesis. This kind of test consists of imposing under the null hypothesis a specific parameter scheme derived from a set of chosen teams, e.g., by using the triadic median. In Tables 13–15 (Appendix D), we perform the OSDM test for each of the proposed reference teams, ranking the teams according to their OSDM estimated statistic and its associated p -value in the three UCL Seasons.

Considering only the first examples for the sake of brevity, European teams that were closer to Real Madrid in a triadic point of view were Barcelona (2017–18 and 2018–19), Juventus (2017–18 and 2018–19) Bayern Munchen (2016–17 and 2018–19), PSG (first two Seasons), but also Naples and Borussia Dortmund (last Season) and Arsenal (first Season). Conversely, in terms of triadic distribution it can be found in Dynamo Zagreb, Rostov, Spartak Moscow, APOEL and PSV.

We would like to note that the results of the OSDM test discussed here are highly similar considering the adjusted type I error rate obtained through the Bonferroni correction. Indeed, observing the Tables in Appendix D, the null hypothesis can be rejected for all the teams such that $p < \alpha/192$, approximately equal to 0.0003 when $\alpha = 0.05$.

To conclude, network structure and triadic analysis can provide information for trainers, coaches and policy-makers in football. Triad census can highlight tactical and technical aspects, and may be able to separate out the cohesion of a team in terms of passing behaviour, clarifying types and the strength of relationships among players, as well as express the collectivity of sub-groups. Therefore, from a tactical perspective, knowledge of opposite triadic distribution may provide a specific improvement both at individual level and for the overall team, also becoming relevant to the outcome of a game, e.g., the probability of winning.

5 Conclusions and future works

Passing networks may provide useful information on the style of play and performance of football teams (Diquigiovanni and Scarpa 2019). In addition, descriptive analysis of dyads and triads is common in the social network analysis field. In particular, triad census is capable of characterizing the network structure, including from a comparative perspective. The present paper exploits the potential of the triad census to analyse networks, interpreting the presence or absence of specific triads as characteristics of different playing styles and/or coaching strategies.

Given some levels of probabilistic assumptions, we showed that it is possible to retrieve a particular triadic configuration in football matches. On such bases, we adopted two types of formal tests. The former was an omnibus test used to determine whether the triadic distribution of teams deviated from randomness. The latter was a DM-based test redesigned to assume a null hypothesis of adherence to a given strategy (in terms of triadic configuration) or to detect similarity between teams.

In our case study, the omnibus test evidenced that there were teams particularly able to create as many links as possible, while other teams showed their strategy to be built on making few connections between players. Moreover, since football teams are always oriented by real-time coach strategies, often the triad census of such teams may be similar to those obtained via random graphs based on the U|MAN assumptions. This means that sometimes the football teams were unable to pursue a given strategy. The reasons for this might lie in several causes, e.g., when the opponent is able to break the playing schemes, when players are not particularly skilful with respect to the opponents, or in the absence of influential players having a key role in sorting passes.

From a slightly different point of view, the OSDM tests help to identify almost three different strategies according to the triadic distribution. One in particular (T_3) was related

to the highly tactical and technical teams, able to reach around 70 % of two types of more connected triads (“300” and “210”). Another strategy (denoted as T_1) seemed to be prerogative of Dutch and French teams, presenting greater balance between types “210” and “201”. A third group (T_2) was mainly composed of teams from Eastern Europe, sharing the types of triad “120” characterized by asymmetric links. As mentioned, this approach can be also used to rank teams according to a specific null hypothesis. We have also tried to use reference teams to see how teams can be close to or far from the triadic behaviour arising from the null hypothesis.

Despite the usefulness of this analysis, the next step would be the development of a weighted triad approach to take into account the different degrees of connections between players involved in a match. In this regard, a further methodology to combine the isomorphism classes of directed networks with the degree of links connecting nodes (e.g., the number of passes between players) should be developed. Moreover, a potential approach to extend the analysis to the weighted cases has been suggested by Onnela et al. (2005), even if applied in a different framework. To the best of our knowledge, a similar approach has not been fully explored in the current literature.

Indeed, passes may have different features according to their length, their position on the pitch (Rein, Raabe, and Memmert 2017), the role of players involved and last but not least, their effectiveness in creating chances (Bransen, Van Haaren, and van de Velden 2019).

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Appendix A: Moments of the U|MAN distribution

In this Section we depict the expected value, the variance and the covariance of the U|MAN distribution:

$$E(T_\eta) = \binom{n}{3} p(\eta); \quad (9)$$

$$\begin{aligned} \text{Var}(T_\eta) &= \binom{n}{3} p(\eta)(1 - p(\eta)) \\ &+ \binom{n}{3}^{(2)} \left(\frac{n+7}{n-2} \right) [p_0(\eta, \eta) - p^2(\eta)] \\ &+ 3 \binom{n}{3} (n-3) [p_2(\eta, \eta) - p^2(\eta)] \end{aligned} \quad (10)$$

$$\begin{aligned} \text{Cov}(T_\eta, T_\nu) &= -\binom{n}{3} p(\eta) p(\nu) \\ &+ \binom{n}{3}^{(2)} \left(\frac{n+7}{n-2} \right) [p_0(\eta, \nu) - p(\eta) p(\nu)] \\ &+ 3 \binom{n}{3} (n-3) [p_2(\eta, \nu) - p(\eta) p(\nu)] \end{aligned} \quad (11)$$

where the set of probabilities $p(\eta)$, $p_0(\eta, \nu)$, $p_2(\eta, \nu)$ are computed in Holland and Leinhardt (1976) (Tables 2–4) and the symbol $\binom{n}{3}^{(2)}$ represents the descending factorial $\binom{n}{3} \left[\binom{n}{3} - 1 \right]$.

Appendix B: The Dirichlet-Multinomial distribution

The DM distribution (also known as Dirichlet-Compound Multinomial, multivariate Binomial-Beta distribution or multivariate Pólya, as defined in Ishii and Hayakawa 1960; Johnson, Kotz, and Balakrishnan 1997), is a family of discrete multivariate probability distributions on a finite support of non-negative integers. Consider a random vector of m distinct modality counts $\mathbf{x} = (x_1, \dots, x_m)$, the density function of a DM distribution with parameters $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m)$ is

$$P(\mathbf{X} = \mathbf{x} | \alpha_1, \dots, \alpha_m) = x_0! \frac{\Gamma(\alpha_0)}{\Gamma(x_0 + \alpha_0)} \prod_{i=1}^m \frac{\Gamma(x_i + \alpha_i)}{\Gamma(\alpha_i) x_i!} \quad (12)$$

where $x_i \in \mathbb{N} \cup \{0\}$, $\min_i(x_i) > 0$ and $\sum_{i=1}^m x_i = x_0$ is the sum of all modalities' counts. Distribution parameters are positive, i.e., $\alpha_i > 0$, and $\sum_{i=1}^m \alpha_i = \alpha_0$.

Alternatively, a different parametrization is given setting $\alpha_0 = (1 - \theta)/\theta$ and $\alpha_i = \pi_i(1 - \theta)/\theta$, with $i = 1, \dots, m$. In this case parameters π_i represent the (theoretical) relative frequencies of each modality, while θ is an overdispersion parameter. The density function in Equation (12) then becomes

$$\begin{aligned} P(\mathbf{X} = \mathbf{x} | \pi_1, \dots, \pi_m, \theta) \\ = \frac{x_0!}{x_1! \dots x_m!} \cdot \frac{\prod_{i=1}^m \prod_{j=1}^{x_i} [\pi_i(1 - \theta) + (j - 1)\theta]}{\prod_{j=1}^{x_0} [(1 - \theta) + (j - 1)\theta]} \end{aligned} \quad (13)$$

Obviously, the same constraints already defined on the support hold, while the new parameter constraints equal to $0 \leq \pi_i \leq 1$, $\theta \geq 0$ and $\sum_{i=1}^m \pi_i = 1$. DM distribution has been formally defined as a compound distribution where a probability vector $\boldsymbol{\pi}$ is assumed to follow a Dirichlet distribution and an observation $\mathbf{x}$ is drawn from a Multinomial distribution with given probability vector $\boldsymbol{\pi}$ and x_0 number of trials.

Using the second parametrization and following Tvedebrink (2010), it is possible to derive mean and variance of the random variable

$$E(\mathbf{X}) = x_0 \boldsymbol{\pi} \quad (14)$$

$$\text{Var}(\mathbf{X}) = x_0 (\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top) (1 + \theta(x_0 - 1)) \quad (15)$$

It is worth to note that distribution parameters do not directly depend on total counts x_0 of all the occurrences present in the sample, but only on the relative frequencies of each modality.

Appendix C: Performance of OSDM test in small samples

Here we perform a small-scale Monte Carlo experiment to obtain the empirical size and power of the OSDM test. We generate 100,000 Monte Carlo samples from a Dirichlet-Multinomial distribution $\mathbf{X} \sim \text{DM}_m(\boldsymbol{\pi}, \theta)$.

We set a fixed $\theta = 0.01$ generating 6 observations having $m = 16$ modalities and a total number of triads for each team equal to 165, mimicking the structure of football data. The parameter scheme under the null hypothesis equals to $\pi_i = 1/16$, for $i = 1, \dots, 16$. To compute the power we let variate the set of probabilities by unbalancing one of the frequencies proportionally to the remaining others, i.e., $\pi_1 =$

Table 8: Size of OSDM test at nominal level $\alpha = 0.05$.

n	$m = 16$
6	0.0766
13	0.0598
38	0.0544
50	0.0524

$1/16 \pm k$, with $k \in \{0, 0.01, 0.02, \dots\}$, and $\pi_h = (1 - \pi_1)/15$, $\forall h = 2, \dots, m$.

In Table 8 the size of the test is computed as function of increasing sample size while the empirical power of the OSDM test is depicted in Figure 11, according to different values of k . The null hypothesis is in correspondence of $k = 0$.

We have to remark that power analysis of OSDM test can be exploited in many ways and here we present only a possible scenario related to changes in a single count. More power analysis will be carried out in further research to take into account the multidimensional nature of the test.

Appendix D: Results of OSDM tests

In what follows we present the results of the OSDM test considering firstly the centroids of k -means after CA, and then using reference teams (i.e., three different styles of play).

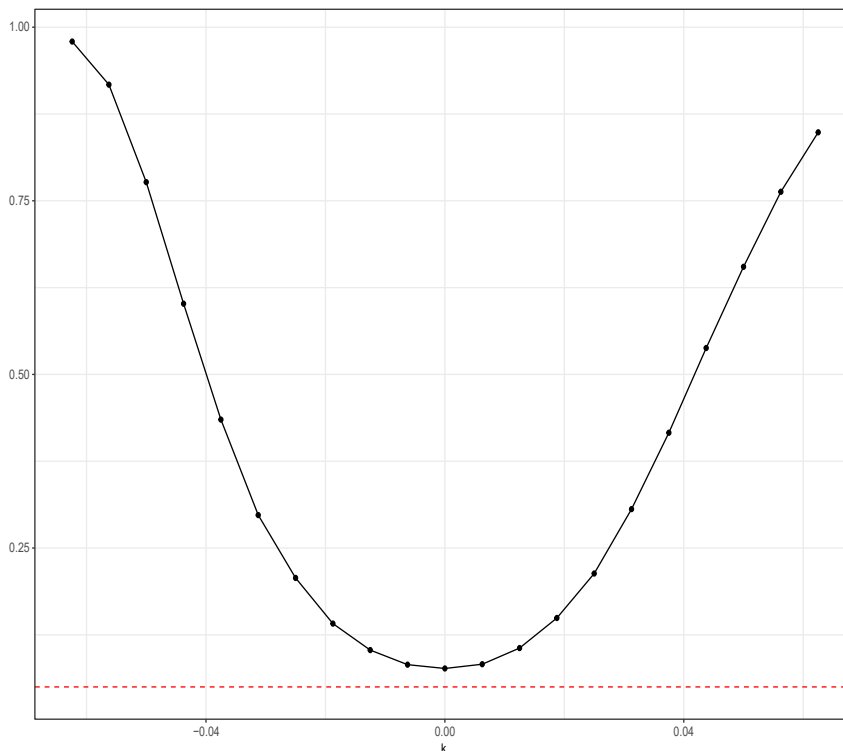


Figure 11: Empirical power of the OSDM test as function of k with $n = 6$ samples.

Table 9: Results of one-sample DM test under $H_0: T_1$.

2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
PSV	2.90 (0.9997)	Leipzig	6.78 (0.9634)	Monaco	6.77 (0.9636)
Dinamo Kiev	3.24 (0.9994)	Napoli	8.69 (0.8932)	Inter	7.29 (0.9492)
Borussia Gladbach	4.40 (0.9961)	Benfica	10.97 (0.7548)	Benfica	8.90 (0.8826)
Monaco	4.55 (0.9953)	Olympiacos	11.28 (0.7322)	Atletico Madrid	9.69 (0.8385)
Lyon	9.05 (0.8750)	Liverpool	12.31 (0.6551)	Lyon	11.40 (0.7236)
Atletico Madrid	9.36 (0.8580)	Celtic	13.97 (0.5282)	Ajax	13.06 (0.5973)
Napoli	10.62 (0.7792)	Feyenord	14.65 (0.4766)	PSG	18.08 (0.2584)
Benfica	11.49 (0.7170)	Anderlecht	15.81 (0.3951)	Liverpool	18.75 (0.2255)
Barcelona	11.66 (0.7046)	Borussia Dortmund	16.12 (0.3741)	Porto	19.09 (0.2096)
Sporting Lisbon	14.13 (0.5159)	Monaco	16.47 (0.3514)	Napoli	19.74 (0.1821)
Sevilla	15.29 (0.4310)	Sevilla	17.55 (0.2872)	Roma	20.92 (0.1394)
Legia Warsav	17.54 (0.2874)	Atletico Madrid	17.79 (0.2738)	Shakhtar Donetsk	21.19 (0.1308)
Manchester City	18.61 (0.2319)	Chelsea	17.82 (0.2723)	Valencia	21.81 (0.1128)
Tottenham	18.79 (0.2233)	CSKA Moskov	19.04 (0.2118)	Viktoria Plzen	23.23 (0.0794)
Ludogorets	21.99 (0.1081)	Porto	20.52 (0.1530)	Manchester United	23.30 (0.0779)
Arsenal	24.44 (0.0580)	Basel	22.81 (0.0883)	Hoffenheim	24.07 (0.0639)
Basel	27.24 (0.0269)	PSG	23.48 (0.0745)	Tottenham	26.40 (0.0340)
Besiktas	30.10 (0.0116)	Sporting Lisbon	25.16 (0.0478)	Schalke 04	26.71 (0.0312)
Brugge	30.97 (0.0089)	Roma	26.71 (0.0312)	Brugge	26.96 (0.0291)
Leicester	31.04 (0.0087)	Manchester City	28.93 (0.0164)	Juventus	30.16 (0.0114)
Juventus	32.60 (0.0053)	Maribor	29.30 (0.0147)	Manchester City	38.01 (0.0009)
CSKA Moskov	38.41 (0.0008)	Manchester United	29.48 (0.0140)	Bayern Munchen	40.64 (0.0004)
Porto	42.90 (0.0002)	Juventus	32.41 (0.0057)	CSKA Moskov	40.97 (0.0003)
PSG	46.35 (0.0000)	Qarabag	36.82 (0.0013)	Barcelona	41.34 (0.0003)
Celtic	54.48 (0.0000)	Besiktas	41.15 (0.0003)	Galatasaray	43.55 (0.0001)
Bayer Leverkusen	70.43 (0.0000)	Barcelona	52.97 (0.0000)	Real Madrid	44.62 (0.0001)
Borussia Dortmund	71.42 (0.0000)	Shakhtar Donetsk	70.90 (0.0000)	AEK Athens	48.64 (0.0000)
Bayern Munchen	74.65 (0.0000)	Tottenham	71.27 (0.0000)	Young Boys	50.40 (0.0000)
Real Madrid	79.34 (0.0000)	APOEL	93.73 (0.0000)	Red Star Belgrade	51.80 (0.0000)
Rostov	110.25 (0.0000)	Real Madrid	97.06 (0.0000)	Borussia Dortmund	55.22 (0.0000)
Copenaghen	114.24 (0.0000)	Spartak Moskov	102.30 (0.0000)	PSV	102.70 (0.0000)
Dynamo Zagreb	117.70 (0.0000)	Bayern Munchen	114.20 (0.0000)	Lokomotiv Moskov	130.26 (0.0000)

p -Values are denoted in brackets.

Table 10: Results of one-sample DM test under $H_0: T_2$.

2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
Copenhagen	7.20 (0.9517)	Besiktas	7.38 (0.9463)	AEK Athens	4.45 (0.9959)
Legia Warsaw	10.09 (0.8138)	Feyenord	8.12 (0.9191)	Galatasaray	8.98 (0.8783)
Bayer Leverkusen	12.72 (0.6239)	Qarabag	10.41 (0.7929)	Hoffenheim	11.33 (0.7286)
Brugge	15.16 (0.4396)	Basel	17.90 (0.2680)	Inter	13.32 (0.5774)
CSKA Moscow	15.25 (0.4338)	Tottenham	22.33 (0.0993)	CSKA Moscow	14.77 (0.4684)
Atletico Madrid	19.65 (0.1858)	Porto	22.35 (0.0989)	Lokomotiv Moscow	14.91 (0.4579)
Leicester	24.19 (0.0619)	Spartak Moscow	22.80 (0.0884)	Porto	18.34 (0.2454)
Borussia Gladbach	30.60 (0.0099)	CSKA Moscow	23.71 (0.0701)	PSG	18.42 (0.2410)
Sevilla	30.94 (0.0089)	Benfica	25.85 (0.0396)	Young Boys	19.11 (0.2086)
Rostov	37.63 (0.0010)	Olympiacos	27.94 (0.0219)	Red Star Belgrade	19.75 (0.1816)
Arsenal	37.86 (0.0009)	Anderlecht	29.89 (0.0123)	PSV	29.71 (0.0130)
Porto	40.11 (0.0004)	Liverpool	33.86 (0.0036)	Shakhtar Donetsk	31.93 (0.0066)
PSV	40.22 (0.0004)	Monaco	37.62 (0.0010)	Brugge	34.79 (0.0026)
Tottenham	42.66 (0.0002)	Roma	41.28 (0.0003)	Tottenham	35.71 (0.0019)
Monaco	44.32 (0.0001)	Leipzig	41.43 (0.0003)	Lyon	40.92 (0.0003)
Ludogorets	47.26 (0.0000)	Maribor	41.73 (0.0002)	Valencia	41.15 (0.0003)
Dinamo Kiev	48.14 (0.0000)	Celtic	44.30 (0.0001)	Monaco	44.43 (0.0001)
Lyon	50.38 (0.0000)	Chelsea	46.66 (0.0000)	Schalke 04	45.51 (0.0001)
Juventus	55.25 (0.0000)	PSG	55.12 (0.0000)	Viktoria Plzen	46.21 (0.0000)
Barcelona	56.30 (0.0000)	Manchester City	58.88 (0.0000)	Napoli	46.85 (0.0000)
Manchester City	56.76 (0.0000)	Juventus	67.04 (0.0000)	Juventus	48.27 (0.0000)
Dynamo Zagreb	58.36 (0.0000)	Borussia Dortmund	67.15 (0.0000)	Liverpool	48.59 (0.0000)
Celtic	60.57 (0.0000)	Sporting Lisbon	68.92 (0.0000)	Bayern Munchen	53.77 (0.0000)
Besiktas	65.48 (0.0000)	Sevilla	69.73 (0.0000)	Barcelona	66.90 (0.0000)
Basel	72.84 (0.0000)	APOEL	72.76 (0.0000)	Atletico Madrid	67.50 (0.0000)
Sporting Lisbon	73.57 (0.0000)	Manchester United	75.49 (0.0000)	Real Madrid	87.80 (0.0000)
PSG	86.32 (0.0000)	Napoli	82.39 (0.0000)	Benfica	91.45 (0.0000)
Benfica	89.62 (0.0000)	Shakhtar Donetsk	82.73 (0.0000)	Roma	93.40 (0.0000)
Napoli	105.42 (0.0000)	Atletico Madrid	86.50 (0.0000)	Manchester United	94.01 (0.0000)
Borussia Dortmund	146.99 (0.0000)	Barcelona	137.66 (0.0000)	Manchester City	95.29 (0.0000)
Real Madrid	153.74 (0.0000)	Real Madrid	190.99 (0.0000)	Borussia Dortmund	109.76 (0.0000)
Bayern Munchen	162.06 (0.0000)	Bayern Munchen	258.83 (0.0000)	Ajax	134.18 (0.0000)

p-Values are denoted in brackets.

Table 11: Results of one-sample DM test under $H_0: T_3$.

2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
PSG	2.68 (0.9998)	Juventus	2.06 (1.0000)	Napoli	2.14 (1.0000)
Borussia Dortmund	3.70 (0.9986)	Manchester City	3.06 (0.9995)	Barcelona	3.43 (0.9991)
Juventus	5.11 (0.9911)	Roma	6.28 (0.9747)	Juventus	3.69 (0.9986)
Tottenham	5.27 (0.9896)	Bayern Munchen	8.50 (0.9020)	Real Madrid	4.67 (0.9946)
Bayern Munchen	5.29 (0.9894)	Barcelona	10.81 (0.7658)	Borussia Dortmund	4.80 (0.9937)
Arsenal	5.50 (0.9870)	Real Madrid	14.93 (0.4564)	Manchester City	7.71 (0.9349)
Real Madrid	8.28 (0.9120)	Sevilla	17.45 (0.2929)	Liverpool	9.98 (0.8212)
Manchester City	8.32 (0.9103)	Benfica	18.55 (0.2350)	Shakhtar Donetsk	11.13 (0.7436)
Sevilla	10.08 (0.8148)	PSG	24.03 (0.0646)	Bayern Munchen	16.61 (0.3425)
Ludogorets	20.66 (0.1481)	Liverpool	24.37 (0.0591)	Tottenham	18.46 (0.2391)
Borussia Gladbach	21.91 (0.1102)	Borussia Dortmund	25.95 (0.0385)	Porto	22.84 (0.0876)
Atletico Madrid	24.22 (0.0614)	Shakhtar Donetsk	41.41 (0.0003)	Roma	26.84 (0.0301)
Lyon	25.92 (0.0388)	Manchester United	43.61 (0.0001)	Inter	33.14 (0.0045)
Barcelona	29.31 (0.0147)	Feyenord	43.76 (0.0001)	PSG	33.90 (0.0035)
Brugge	37.15 (0.0012)	Chelsea	47.27 (0.0000)	Atletico Madrid	40.36 (0.0004)
Besiktas	37.70 (0.0010)	Besiktas	56.04 (0.0000)	Lyon	59.08 (0.0000)
Monaco	45.24 (0.0001)	Leipzig	62.26 (0.0000)	Manchester United	62.69 (0.0000)
PSV	46.38 (0.0000)	Anderlecht	65.46 (0.0000)	Valencia	64.86 (0.0000)
Legia Warsaw	50.04 (0.0000)	Napoli	65.58 (0.0000)	Galatasaray	68.41 (0.0000)
Porto	55.70 (0.0000)	Olympiacos	68.16 (0.0000)	Monaco	71.19 (0.0000)
Sporting Lisbon	55.90 (0.0000)	Tottenham	69.82 (0.0000)	Ajax	75.38 (0.0000)
Dinamo Kiev	59.18 (0.0000)	Atletico Madrid	83.00 (0.0000)	Hoffenheim	81.63 (0.0000)
Bayer Leverkusen	82.10 (0.0000)	Basel	85.56 (0.0000)	Young Boys	88.17 (0.0000)
CSKA Moskov	90.91 (0.0000)	Porto	88.41 (0.0000)	AEK Athens	93.05 (0.0000)
Benfica	100.59 (0.0000)	Qarabag	91.22 (0.0000)	CSKA Moskov	99.54 (0.0000)
Leicester	105.51 (0.0000)	CSKA Moskov	91.60 (0.0000)	Viktoria Plzen	106.74 (0.0000)
Napoli	107.73 (0.0000)	Celtic	102.58 (0.0000)	Brugge	110.05 (0.0000)
Basel	151.29 (0.0000)	Monaco	116.41 (0.0000)	Benfica	124.34 (0.0000)
Celtic	184.54 (0.0000)	Maribor	146.27 (0.0000)	Red Star Belgrade	128.87 (0.0000)
Rostov	186.58 (0.0000)	Sporting Lisbon	179.63 (0.0000)	Schalke 04	133.93 (0.0000)
Copenhagen	202.83 (0.0000)	APOEL	258.55 (0.0000)	Lokomotiv Moskov	150.41 (0.0000)
Dynamo Zagreb	263.87 (0.0000)	Spartak Moskov	271.92 (0.0000)	PSV	230.21 (0.0000)

p -Values are denoted in brackets.

Table 12: Results of one-sample DM test under $H_0: T_4$.

2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
Leicester	26.28 (0.0352)	APOEL	17.35 (0.2985)	Schalke 04	18.32 (0.2463)
Dynamo Zagreb	31.01 (0.0088)	Anderlecht	19.97 (0.1731)	Inter	31.55 (0.0074)
Monaco	31.49 (0.0076)	Olympiacos	22.32 (0.0998)	AEK Athens	46.75 (0.0000)
Celtic	43.84 (0.0001)	Celtic	27.66 (0.0238)	Red Star Belgrade	47.71 (0.0000)
Rostov	52.03 (0.0000)	Maribor	30.90 (0.0090)	Monaco	54.74 (0.0000)
Atletico Madrid	53.27 (0.0000)	Monaco	38.29 (0.0008)	Hoffenheim	62.18 (0.0000)
Dinamo Kiev	57.89 (0.0000)	CSKA Moskov	39.39 (0.0006)	Brugge	62.55 (0.0000)
PSV	58.54 (0.0000)	Feyenord	53.03 (0.0000)	PSV	65.40 (0.0000)
Legia Warsav	63.56 (0.0000)	Porto	56.74 (0.0000)	CSKA Moskov	76.38 (0.0000)
Benfica	70.84 (0.0000)	Sporting Lisbon	68.47 (0.0000)	Valencia	80.34 (0.0000)
Borussia Gladbach	75.36 (0.0000)	Besiktas	71.98 (0.0000)	Viktoria Plzen	86.01 (0.0000)
CSKA Moskov	93.72 (0.0000)	Qarabag	76.53 (0.0000)	Galatasaray	89.42 (0.0000)
Sevilla	95.16 (0.0000)	Leipzig	81.97 (0.0000)	Porto	91.48 (0.0000)
Brugge	96.13 (0.0000)	Basel	86.47 (0.0000)	PSG	98.24 (0.0000)
Lyon	101.87 (0.0000)	Benfica	93.99 (0.0000)	Lyon	109.25 (0.0000)
Manchester City	111.22 (0.0000)	Liverpool	98.82 (0.0000)	Atletico Madrid	120.76 (0.0000)
Arsenal	114.47 (0.0000)	Roma	140.50 (0.0000)	Napoli	126.37 (0.0000)
Tottenham	120.53 (0.0000)	Manchester City	146.49 (0.0000)	Shakhtar Donetsk	126.71 (0.0000)
Barcelona	122.53 (0.0000)	Napoli	157.84 (0.0000)	Young Boys	133.78 (0.0000)
Sporting Lisbon	132.04 (0.0000)	Sevilla	163.36 (0.0000)	Tottenham	142.23 (0.0000)
Napoli	135.86 (0.0000)	PSG	164.03 (0.0000)	Liverpool	145.27 (0.0000)
Bayer Leverkusen	136.11 (0.0000)	Chelsea	171.72 (0.0000)	Juventus	148.86 (0.0000)
Ludogorets	163.35 (0.0000)	Tottenham	171.81 (0.0000)	Benfica	151.94 (0.0000)
Basel	170.56 (0.0000)	Atletico Madrid	173.45 (0.0000)	Barcelona	177.27 (0.0000)
Copenhagen	171.08 (0.0000)	Juventus	175.01 (0.0000)	Lokomotiv Moskov	198.41 (0.0000)
Juventus	176.31 (0.0000)	Spartak Moskov	176.03 (0.0000)	Bayern Munchen	201.63 (0.0000)
Porto	178.25 (0.0000)	Borussia Dortmund	179.32 (0.0000)	Manchester City	205.81 (0.0000)
PSG	247.50 (0.0000)	Manchester United	245.07 (0.0000)	Roma	215.06 (0.0000)
Besiktas	251.22 (0.0000)	Barcelona	289.00 (0.0000)	Ajax	224.92 (0.0000)
Real Madrid	349.35 (0.0000)	Shakhtar Donetsk	355.26 (0.0000)	Real Madrid	230.40 (0.0000)
Bayern Munchen	389.47 (0.0000)	Real Madrid	428.29 (0.0000)	Manchester United	250.30 (0.0000)
Borussia Dortmund	406.26 (0.0000)	Bayern Munchen	666.18 (0.0000)	Borussia Dortmund	259.83 (0.0000)

p-Values are denoted in brackets.

Table 13: One-sample test statistics of teams in each season, whose null hypothesis is “structural similarity” Tier A versus reference teams.

Reference: <i>Real Madrid</i>		Reference: <i>Real Madrid</i>		Reference: <i>Real Madrid</i>	
2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
PSG	11.76 (0.6974)	Manchester City	16.24 (0.3663)	Barcelona	6.94 (0.9594)
Manchester City	12.84 (0.6147)	Juventus	17.66 (0.2809)	Napoli	7.05 (0.9562)
Tottenham	17.72 (0.2778)	Barcelona	18.59 (0.2331)	Borussia Dortmund	8.13 (0.9185)
Bayern Munchen	18.94 (0.2165)	PSG	52.70 (0.0000)	Manchester City	9.32 (0.8603)
Arsenal	20.73 (0.1457)	Sevilla	52.97 (0.0000)	Juventus	16.29 (0.3628)
Juventus	27.60 (0.0242)	Roma	59.67 (0.0000)	Bayern Munchen	26.83 (0.0301)
Borussia Dortmund	32.37 (0.0057)	Liverpool	62.89 (0.0000)	Liverpool	28.22 (0.0202)
Sevilla	38.32 (0.0000)	Bayern Munchen	65.16 (0.0000)	Shakhtar Donetsk	34.10 (0.0033)
Lyon	51.50 (0.0000)	Borussia Dortmund	85.44 (0.0000)	Tottenham	42.01 (0.0000)
Borussia Gladbach	58.03 (0.0000)	Benfica	98.06 (0.0000)	Porto	43.89 (0.0001)
Barcelona	76.13 (0.0000)	Feyenord	161.81 (0.0000)	Inter	59.44 (0.0000)
Monaco	88.22 (0.0000)	Leipzig	163.98 (0.0000)	Roma	62.00 (0.0000)
PSV	97.54 (0.0000)	Napoli	164.59 (0.0000)	PSG	67.36 (0.0000)
Atletico Madrid	103.76 (0.0000)	Shakhtar Donetsk	170.90 (0.0000)	Atletico Madrid	68.07 (0.0000)
Brugge	103.79 (0.0000)	Anderlecht	201.17 (0.0000)	Lyon	101.22 (0.0000)
Ludogorets	108.18 (0.0000)	Chelsea	204.27 (0.0000)	Valencia	121.67 (0.0000)
Besiktas	122.51 (0.0000)	Manchester United	205.85 (0.0000)	Ajax	127.00 (0.0000)
Dinamo Kiev	127.17 (0.0000)	Besiktas	206.67 (0.0000)	Monaco	131.60 (0.0000)
Porto	127.81 (0.0000)	Tottenham	207.34 (0.0000)	Benfica	151.22 (0.0000)
Legia Warsav	132.25 (0.0000)	Atletico Madrid	214.07 (0.0000)	Galatasaray	159.68 (0.0000)
Sporting Lisbon	151.47 (0.0000)	Olympiacos	220.13 (0.0000)	Manchester United	164.68 (0.0000)
Benfica	213.55 (0.0000)	Basel	239.46 (0.0000)	CSKA Moskov	176.33 (0.0000)
Bayer Leverkusen	227.12 (0.0000)	Celtic	265.64 (0.0000)	Brugge	183.50 (0.0000)
CSKA Moskov	229.24 (0.0000)	Porto	286.50 (0.0000)	Viktoria Plzen	195.40 (0.0000)
Napoli	270.75 (0.0000)	Monaco	313.33 (0.0000)	AEK Athens	195.65 (0.0000)
Leicester	276.03 (0.0000)	Qarabag	315.74 (0.0000)	Hoffenheim	201.17 (0.0000)
Basel	400.26 (0.0000)	CSKA Moskov	333.67 (0.0000)	Young Boys	217.99 (0.0000)
Copenhagen	534.15 (0.0000)	Maribor	407.93 (0.0000)	Schalke 04	289.88 (0.0000)
Celtic	537.47 (0.0000)	Sporting Lisbon	590.96 (0.0000)	Lokomotiv Moskov	303.20 (0.0000)
Rostov	645.08 (0.0000)	APOEL	836.93 (0.0000)	Red Star Belgrade	354.21 (0.0000)
Dynamo Zagreb	794.94 (0.0000)	Spartak Moskov	843.63 (0.0000)	PSV	511.75 (0.0000)

p -Values are denoted in brackets.

Table 14: One-sample test statistics of teams in each season, whose null hypothesis is “structural similarity” of Tier B versus reference teams.

Reference: <i>Sporting Lisbon</i>		Reference: <i>Sporting Lisbon</i>		Reference: <i>Benfica</i>	
2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
Lyon	8.32 (0.9105)	Monaco	3.82 (0.9983)	Inter	8.14 (0.9180)
Borussia Gladbach	15.03 (0.4494)	Olympiacos	4.03 (0.9976)	Atletico Madrid	14.83 (0.4637)
Tottenham	15.90 (0.3890)	Celtic	4.40 (0.9961)	PSG	15.47 (0.4181)
Manchester City	17.84 (0.2714)	CSKA Moskov	5.98 (0.9800)	Porto	17.54 (0.2876)
Sevilla	17.88 (0.2692)	Porto	6.72 (0.9649)	Lyon	17.55 (0.2870)
Arsenal	21.74 (0.1148)	Anderlecht	6.91 (0.9601)	Liverpool	20.82 (0.1428)
Atletico Madrid	24.99 (0.0500)	Feyenord	9.22 (0.8658)	Shakhtar Donetsk	21.63 (0.1180)
Ludogorets	25.79 (0.0403)	Maribor	11.04 (0.7498)	Napoli	22.72 (0.0903)
Barcelona	33.40 (0.0041)	Leipzig	13.36 (0.5747)	Monaco	24.48 (0.0574)
Monaco	33.92 (0.0035)	Qarabag	16.76 (0.3334)	Roma	25.67 (0.0416)
Juventus	37.13 (0.0012)	Basel	17.33 (0.2995)	Brugge	26.24 (0.0356)
Dinamo Kiev	39.86 (0.0005)	Benfica	21.71 (0.1155)	Tottenham	26.79 (0.0305)
PSG	41.89 (0.0002)	Liverpool	25.91 (0.0390)	Viktoria Plzen	29.64 (0.0133)
PSV	43.26 (0.0001)	Besiktas	28.54 (0.0184)	Ajax	30.82 (0.0093)
Besiktas	45.84 (0.0001)	Napoli	33.56 (0.0039)	Valencia	30.86 (0.0092)
Benfica	46.71 (0.0000)	Chelsea	35.76 (0.0019)	Juventus	33.04 (0.0046)
Legia Warsaw	52.61 (0.0000)	Atletico Madrid	40.96 (0.0003)	Hoffenheim	33.41 (0.0041)
Porto	55.79 (0.0000)	Borussia Dortmund	42.65 (0.0002)	Bayern Munchen	37.07 (0.0012)
Borussia Dortmund	57.86 (0.0000)	Roma	45.00 (0.0001)	Manchester United	41.28 (0.0003)
Bayer Leverkusen	60.17 (0.0000)	PSG	46.18 (0.0000)	Galatasaray	42.06 (0.0002)
Bayern Munchen	62.45 (0.0000)	Sevilla	48.68 (0.0000)	Schalke 04	44.02 (0.0001)
Real Madrid	65.15 (0.0000)	APOEL	51.15 (0.0000)	Manchester City	45.34 (0.0001)
Napoli	69.69 (0.0000)	Spartak Moskov	52.46 (0.0000)	Barcelona	45.43 (0.0001)
CSKA Moskov	85.48 (0.0000)	Manchester City	54.52 (0.0000)	Real Madrid	47.13 (0.0000)
Brugge	99.05 (0.0000)	Manchester United	59.39 (0.0000)	AEK Athens	47.30 (0.0000)
Celtic	154.96 (0.0000)	Tottenham	61.62 (0.0000)	CSKA Moskov	50.95 (0.0000)
Rostov	165.26 (0.0000)	Juventus	61.78 (0.0000)	Young Boys	58.85 (0.0000)
Leicester	176.68 (0.0000)	Shakhtar Donetsk	102.31 (0.0000)	Borussia Dortmund	62.78 (0.0000)
Basel	234.35 (0.0000)	Barcelona	107.65 (0.0000)	Red Star Belgrade	69.09 (0.0000)
Copenhagen	268.88 (0.0000)	Real Madrid	175.79 (0.0000)	PSV	119.95 (0.0000)
Dynamo Zagreb	563.09 (0.0000)	Bayern Munchen	227.70 (0.0000)	Lokomotiv Moskov	125.42 (0.0000)

p -Values are denoted in brackets.

Table 15: One-sample test statistics of teams in each season, whose null hypothesis is “structural similarity” of Tier C versus reference teams.

Reference: <i>Dynamo Zagreb</i>		Reference: <i>Anderlecht</i>		Reference: <i>Monaco</i>	
2016–2017	X_{obs}	2017–2018	X_{obs}	2018–2019	X_{obs}
Rostov	15.99 (0.3828)	Olympiacos	6.84 (0.9620)	Inter	6.84 (0.9618)
Leicester	16.30 (0.3621)	Monaco	15.08 (0.4460)	Viktoria Plzen	9.55 (0.8469)
Celtic	36.03 (0.0017)	CSKA Moskov	17.45 (0.2928)	Brugge	13.38 (0.5730)
Legia Warsav	38.70 (0.0007)	Celtic	17.89 (0.2686)	Benfica	15.33 (0.4282)
Monaco	42.34 (0.0002)	Porto	18.85 (0.2208)	Lyon	15.92 (0.3875)
Atletico Madrid	46.61 (0.0000)	Feyenord	21.02 (0.1361)	PSG	16.43 (0.3539)
CSKA Moskov	57.08 (0.0000)	Liverpool	21.37 (0.1255)	Hoffenheim	16.69 (0.3376)
Dinamo Kiev	61.37 (0.0000)	Leipzig	23.47 (0.0747)	Atletico Madrid	17.09 (0.3133)
PSV	63.31 (0.0000)	Maribor	24.05 (0.0643)	Porto	18.14 (0.2553)
Brugge	66.44 (0.0000)	Benfica	24.78 (0.0530)	Schalke 04	21.83 (0.1125)
Borussia Gladbach	69.95 (0.0000)	Qarabag	29.38 (0.0144)	Valencia	22.94 (0.0853)
Copenhagen	73.85 (0.0000)	Basel	34.24 (0.0032)	CSKA Moskov	23.76 (0.0693)
Bayer Leverkusen	82.11 (0.0000)	Sporting Lisbon	34.44 (0.0030)	Liverpool	24.90 (0.0513)
Sevilla	87.83 (0.0000)	Atletico Madrid	37.11 (0.0012)	Shakhtar Donetsk	25.57 (0.0428)
Benfica	96.79 (0.0000)	Chelsea	38.71 (0.0007)	AEK Athens	28.56 (0.0183)
Lyon	100.31 (0.0000)	Sevilla	41.04 (0.0003)	Tottenham	28.69 (0.0176)
Arsenal	102.89 (0.0000)	Roma	42.79 (0.0002)	Napoli	29.39 (0.0143)
Tottenham	110.55 (0.0000)	PSG	43.07 (0.0002)	Galatasaray	30.48 (0.0103)
Manchester City	112.70 (0.0000)	APOEL	43.11 (0.0002)	Ajax	31.90 (0.0066)
Barcelona	113.59 (0.0000)	Napoli	44.41 (0.0001)	Roma	34.77 (0.0027)
Sporting Lisbon	135.15 (0.0000)	Besiktas	45.52 (0.0001)	Juventus	38.30 (0.0008)
Porto	136.46 (0.0000)	Manchester City	46.83 (0.0000)	Red Star Belgrade	38.44 (0.0008)
Ludogorets	140.47 (0.0000)	Borussia Dortmund	51.75 (0.0000)	Young Boys	39.86 (0.0005)
Basel	145.57 (0.0000)	Juventus	56.60 (0.0000)	Manchester United	42.33 (0.0002)
Juventus	153.92 (0.0000)	Tottenham	71.39 (0.0000)	Bayern Munchen	44.89 (0.0001)
Napoli	155.98 (0.0000)	Spartak Moskov	76.23 (0.0000)	Barcelona	53.06 (0.0000)
Besiktas	208.52 (0.0000)	Manchester United	79.70 (0.0000)	Manchester City	55.06 (0.0000)
PSG	224.55 (0.0000)	Barcelona	94.60 (0.0000)	Real Madrid	60.23 (0.0000)
Real Madrid	339.19 (0.0000)	Shakhtar Donetsk	114.47 (0.0000)	Borussia Dortmund	74.86 (0.0000)
Borussia Dortmund	372.93 (0.0000)	Real Madrid	137.29 (0.0000)	PSV	93.66 (0.0000)
Bayern Munchen	375.26 (0.0000)	Bayern Munchen	208.59 (0.0000)	Lokomotiv Moskov	99.75 (0.0000)

p -Values are denoted in brackets.

References

- Albert, J., Y. Bennett, and J. J. Cochran. 2005. *Anthology of Statistics in Sports*. United States: SIAM.
- Bate, R. 1988. "Football Chance: Tactics and Strategy." In *Science and Football: Proceedings of the First World Congress of Science and Football*, edited by T. Reilly, A. Lees, and K. Davids. United States: Routledge.
- Baumer, B. S., S. T. Jensen, and G. J. Matthews. 2015. "Openwar: An Open Source System for Evaluating Overall Player Performance in Major League Baseball." *Journal of Quantitative Analysis in Sports* 11 (2): 69–84.
- Block, P. 2015. "Reciprocity, Transitivity, and the Mysterious Three-Cycle." *Social Networks* 40: 163–73.
- Boyle, R., and R. Haynes. 2004. *Football in the New Media Age*. Routledge.
- Bransen, L., J. Van Haaren, and M. van de Velden. 2019. "Measuring Soccer Players' Contributions to Chance Creation by Valuing Their Passes." *Journal of Quantitative Analysis in Sports* 15 (2): 97–116.
- Brier, S. S. 1980. "Analysis of Contingency Tables under Cluster Sampling." *Biometrika* 67 (3): 591–6.
- Buchheit, M., A. Allen, T. K. Poon, M. Modonutti, W. Gregson, and V. Di Salvo. 2014. "Integrating Different Tracking Systems in Football: Multiple Camera Semi-Automatic System, Local Position Measurement and Gps Technologies." *Journal of Sports Sciences* 32 (20): 1844–57.
- Buchheit, M., and B. M. Simpson. 2017. "Player-Tracking Technology: Half-Full or Half-Empty Glass?" *International Journal of Sports Physiology and Performance* 12 (s2): S2–35.
- Caicedo-Parada, S., C. Lago-Peñas, and E. Ortega-Toro. 2020. "Passing Networks and Tactical Action in Football: A Systematic Review." *International Journal of Environmental Research and Public Health* 17 (18): 6649.
- Carpita, M., E. Ciavolino, and P. Pasca. 2019. "Exploring and Modelling Team Performances of the Kaggle European Soccer Database." *Statistical Modelling* 19 (1): 74–101.
- Cintia, P., S. Rinzivillo, and L. Pappalardo. 2015. "A Network-Based Approach to Evaluate the Performance of Football Teams." In *Machine Learning and Data Mining for Sports Analytics Workshop*.
- Clemente, F. M., F. M. L. Martins, and R. S. Mendes. 2016. *Social Network Analysis Applied to Team Sports Analysis*. United States: Springer.
- Clemente, F. M., H. Sarmento, and R. Aquino. 2020. "Player Position Relationships with Centrality in the Passing Network of World Cup Soccer Teams: Win/Loss Match Comparisons." *Chaos, Solitons & Fractals* 133: 109625.
- Cox, D., and F. R. T. Trevor. 2002. "An Analysis of Decathlon Data." *Journal of the Royal Statistical Society: Series D (The Statistician)* 51 (2): 179–87.
- Csardi, G., and T. Nepusz. 2006. *The igraph Software Package for Complex Network Research*, 1695. United States: InterJournal Complex Systems.
- Cugmas, M., A. Ferligoj, and A. Žiberna. 2017. "Generating Global Network Structures by Triad Types." arXiv preprint arXiv:1710.10042.
- Davis, J. A., and S. Leinhardt. 1967. "The Structure of Positive Interpersonal Relations in Small Groups." In *Schlüsselwerke der Netzwerkforschung*. Germany: Springer.
- Diquigiovanni, J., and B. Scarpa. 2019. "Analysis of Association Football Playing Styles: An Innovative Method to Cluster Networks." *Statistical Modelling* 19 (1): 28–54.
- Edgecomb, S., and K. Norton. 2006. "Comparison of Global Positioning and Computer-Based Tracking Systems for Measuring Player Movement Distance During Australian Football." *Journal of Science and Medicine in Sport* 9 (1–2): 25–32.
- Ennis, D. M., and J. Bi. 1999. "The Dirichlet-Multinomial Model: Accounting for Inter-Trial Variation in Replicated Ratings." *Journal of Sensory Studies* 14 (3): 321–45.
- Faust, K. 2006. "Comparing Social Networks: Size, Density, and Local Structure." *Metodoloski zvezki* 3 (2): 185.
- Faust, K. 2007. "7. Very Local Structure in Social Networks." *Sociological Methodology* 37 (1): 209–56.
- Faust, K. 2008. "Triadic Configurations in Limited Choice Sociometric Networks: Empirical and Theoretical Results." *Social Networks* 30 (4): 273–82.
- Faust, K. 2010. "A Puzzle Concerning Triads in Social Networks: Graph Constraints and the Triad Census." *Social Networks* 32 (3): 221–33.
- Fewell, J. H., D. Armbruster, J. Ingraham, A. Petersen, and J. S. Waters. 2012. "Basketball Teams as Strategic Networks." *PLoS One* 7 (11): e47445.
- Gallagher, S. K., K. Frisoli, and A. Luby. 2021. "Opening up the Court: Analyzing Player Performance across Tennis Grand Slams." *Journal of Quantitative Analysis in Sports* 17 (4): 255–71.
- Gomez, R. 2002. "Salary Compression and Team Performance: Evidence from the National Hockey League." In *Sportökonomie*, 203–20. Germany: Springer.
- Gonçalves, B., D. Coutinho, S. Santos, C. Lago-Penas, S. Jiménez, and J. Sampaio. 2017. "Exploring Team Passing Networks and Player Movement Dynamics in Youth Association Football." *PLoS One* 12 (1): e0171156.
- Greenacre, M. J. 1984. *Correspondence Analysis*. London: Academic Press.
- Groll, A., C. Ley, G. Schaubberger, and H. Van Eetvelde. 2019. "A Hybrid Random Forest to Predict Soccer Matches in International Tournaments." *Journal of Quantitative Analysis in Sports* 15 (4): 271–87.
- Groll, A., G. Schaubberger, and G. Tutz. 2015. "Prediction of Major International Soccer Tournaments Based on Team-specific Regularized Poisson Regression: An Application to the Fifa World Cup 2014." *Journal of Quantitative Analysis in Sports* 11 (2): 97–115.
- Grund, T. U. 2012. "Network Structure and Team Performance: The Case of English Premier League Soccer Teams." *Social Networks* 34 (4): 682–90.
- Hadley, L., M. Poitras, J. Ruggiero, and S. Knowles. 2000. "Performance Evaluation of National Football League Teams." *Managerial and Decision Economics* 21 (2): 63–70.
- Holland, P. W., and S. Leinhardt. 1976. "Local Structure in Social Networks." *Sociological Methodology* 7: 1–45.
- Holland, P. W., and S. Leinhardt. 1978. "An Omnibus Test for Social Structure Using Triads." *Sociological Methods & Research* 7 (2): 227–56.
- Hyballa, P., and H.-D. Te Poel. 2015. *German Soccer Passing Drills: More than 100 Drills from the Pros*. Germany: Meyer & Meyer Verlag.
- Ibáñez, S. J., J. Sampaio, S. Feu, A. Lorenzo, M. A. Gómez, and E. Ortega. 2008. "Basketball Game-Related Statistics that Discriminate Between Teams' Season-Long Success." *European Journal of Sport Science* 8 (6): 369–72.
- Ievoli, R., A. Gardini, and L. Palazzo. 2021a. "The Role of Passing Network Indicators in Modeling Football Outcomes: An Application Using Bayesian Hierarchical Models." *ASIA Advances in Statistical Analysis*, 107: 1–23.

- Ievoli, R., L. Palazzo, and G. Ragozini. 2021b. "On the Use of Passing Network Indicators to Predict Football Outcomes." *Knowledge-Based Systems* 222: 106997.
- Ishii, G., and R. Hayakawa. 1960. "On the Compound Binomial Distribution." *Annals of the Institute of Statistical Mathematics* 12 (1): 69–80.
- Johnson, N. L., S. Kotz, and N. Balakrishnan. 1997. *Discrete Multivariate Distributions*, Vol. 165. New York: Wiley.
- Kahn, L. M. 1993. "Managerial Quality, Team Success, and Individual Player Performance in Major League Baseball." *ILR Review* 46 (3): 531–47.
- Kaiser, H. F. 1960. "The Application of Electronic Computers to Factor Analysis." *Educational and Psychological Measurement* 20 (1): 141–51.
- Keller, J. B. 1994. "A Characterization of the Poisson Distribution and the Probability of Winning a Game." *The American Statistician* 48 (4): 294–8.
- Kimber, A. C., and A. R. Hansford. 1993. "A Statistical Analysis of Batting in Cricket." *Journal of the Royal Statistical Society: Series A* 156 (3): 443–55.
- Koehler, K. J., and J. R. Wilson. 1986. "Chi-Square Tests for Comparing Vectors of Proportions for Several Cluster Samples." *Communications in Statistics - Theory and Methods* 15 (10): 2977–90.
- Krackhardt, D. 1999. "The Ties that Torture: Simmelian Tie Analysis in Organizations." *Research in the Sociology of Organizations* 16 (1): 183–210.
- La Rosa, P. S., J. P. Brooks, E. Deych, E. L. Boone, D. J. Edwards, Q. Wang, E. Sodergren, G. Weinstock, and W. D. Shannon. 2012. "Hypothesis Testing and Power Calculations for Taxonomic-Based Human Microbiome Data." *PLoS One* 7 (12): e52078.
- La Rosa, P. S., E. Deych, S. Carter, B. Shands, D. Yang, and W. D. Shannon. 2019. *HMP: Hypothesis Testing And Power Calculations for Comparing Metagenomic Samples from HMP*. R package version 2.0.1.
- Lago-Peñas, C., and A. Dellal. 2010. "Ball Possession Strategies in Elite Soccer According to the Evolution of the Match-Score: The Influence of Situational Variables." *Journal of Human Kinetics* 25: 93–100.
- Lewis, A. 2005. "Towards Fairer Measures of Player Performance in One-Day Cricket." *Journal of the Operational Research Society* 56 (7): 804–15.
- Lorrain, F., and H. C. White. 1971. "Structural Equivalence of Individuals in Social Networks." *Journal of Mathematical Sociology* 1 (1): 49–80.
- Luczkovich, J. J., S. P. Borgatti, J. C. Johnson, and M. G. Everett. 2003. "Defining and Measuring Trophic Role Similarity in Food Webs Using Regular Equivalence." *Journal of Theoretical Biology* 220 (3): 303–21.
- McIntyre, S., and R. McKittrick. 2005. "Hockey Sticks, Principal Components, and Spurious Significance." *Geophysical Research Letters* 32 (3): 1–6.
- McClean, S., P. M. Salmon, A. D. Gorman, N. J. Stevens, and C. Solomon. 2018. "A Social Network Analysis of the Goal Scoring Passing Networks of the 2016 European Football Championships." *Human Movement Science* 57: 400–8.
- Mosimann, J. E. 1962. "On the Compound Multinomial Distribution, the Multivariate β -Distribution, and Correlations Among Proportions." *Biometrika* 49 (1/2): 65–82.
- Minka, T. 2000. *Estimating a Dirichlet Distribution*. <https://tminka.github.io/papers/dirichlet/>.
- Moura, F. A., L. E. B. Martins, and S. A. Cunha. 2014. "Analysis of Football Game-Related Statistics Using Multivariate Techniques." *Journal of Sports Sciences* 32 (20): 1881–7.
- Onnela, J.-P., J. Saramäki, J. Kertész, and K. Kaski. 2005. "Intensity and Coherence of Motifs in Weighted Complex Networks." *Physical Review E* 71 (6): 065103.
- Pina, T. J., A. Paulo, and D. Araújo. 2017. "Network Characteristics of Successful Performance in Association Football. A Study on the Uefa Champions League." *Frontiers in Psychology* 8: 1173.
- Rein, R., D. Raabe, and D. Memmert. 2017. "Which Pass Is Better? Novel Approaches to Assess Passing Effectiveness in Elite Soccer." *Human Movement Science* 55: 172–81.
- Ricard, I., and A. Davison. 2007. "Statistical Inference for Olfactometer Data." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 56 (4): 479–92.
- Sampaio, J., M. Janeira, S. Ibáñez, and A. Lorenzo. 2006. "Discriminant Analysis of Game-Related Statistics between Basketball Guards, Forwards and Centres in Three Professional Leagues." *European Journal of Sport Science* 6 (3): 173–8.
- Stern, H. 1991. "On the Probability of Winning a Football Game." *The American Statistician* 45 (3): 179–83.
- Tibshirani, R. J., A. Price, and J. Taylor. 2011. "A Statistician Plays Darts." *Journal of the Royal Statistical Society: Series A* 174 (1): 213–26.
- Tvedebrink, T. 2010. "Overdispersion in Allelic Counts and θ -Correction in Forensic Genetics." *Theoretical Population Biology* 78 (3): 200–10.
- Wasserman, S., and K. Faust. 1994. *Social Network Analysis: Methods and Applications*, 8. United Kingdom: Cambridge University Press.
- Wu, S. H., R. S. Schwartz, D. J. Winter, D. F. Conrad, and R. A. Cartwright. 2017. "Estimating Error Models for Whole Genome Sequencing Using Mixtures of Dirichlet-Multinomial Distributions." *Bioinformatics* 33 (15): 2322–9.
- Yurko, R., S. Ventura, and M. Horowitz. 2019. "Nflwar: A Reproducible Method for Offensive Player Evaluation in Football." *Journal of Quantitative Analysis in Sports* 15 (3): 163–83.